Chapter 23

Class II Cast Metal Restorations

"You arrive at precision…
When you become precise in your technique…"
—Anonymous

The cast metal restoration is an *indirect restoration* that involves numerous steps and dental materials, with meticulous attention to detail. Typically, a dental laboratory is involved, and the dentist and the laboratory technician must be devoted to perfection. The high degree of satisfaction and service derived from a properly made cast metal restoration is a reward for the painstaking application required.[1]

- The class II *inlay* is an intra-coronal cast metal restoration that involves the occlusal and proximal surfaces of a posterior tooth.
- The *partial onlay* is a cast metal restoration that involves the occlusal and proximal surfaces of a posterior and covers and restores *at least one* but not all of the cusp tips of a posterior tooth.
- The class II *onlay* is a cast metal restoration that involves the occlusal and proximal surfaces of a posterior tooth and caps *all of the cusps*.

Cast Metal Alloys

Cast metal restorations can be made from a variety of casting alloys. Their high compressive and tensile strengths are especially valuable in restorations that rebuild *most* or *all* of the occlusal surface.

At present, four distinct groups of alloys are in use for cast restorations:

I. Traditional high-gold alloys (*ADA specification No. 5*)
II. Low-gold alloys
III. Palladium–silver alloys
IV. Base metal alloys (*most commonly used*).

The American Dental Association (ADA) Specification No. 5 for dental casting gold alloys requires a minimum total gold-plus-noble-metals content of 75 wt%. Such traditional high-gold alloys are unreactive in the oral environment and are some of the most biocompatible materials available to the restorative dentist.[2] However, their usage has drastically reduced due to the increasing cost of gold and other noble alloys.

- Each of the alternatives to high-gold alloys have associated problems of reduced performance, most commonly related to:
  - Decreased tarnish resistance.
  - Decreased burnishability.[3]
  - Higher incidences of postrestorative allergy, most often exhibited by irritated soft tissue adjacent to the restoration.

Indications

Large Restorations

I. When proximal surface caries is extensive, the cast metal inlay is an alternative to amalgam or composite when the higher strength of a casting alloy is needed.
II. The cast metal onlay is often an excellent alternative to a crown for teeth that have been greatly weakened by caries or by large, failing restorations but where the facial and lingual tooth surfaces are relatively unaffected by disease or injury.

Superior Contact and Contour

The indirect procedure used to develop the cast restoration allows more control of contours and contacts (proximal and occlusal).

Endodontically Treated Teeth

A molar or premolar with endodontic treatment can be restored with a cast metal onlay, provided that the onlay has been thoughtfully designed to distribute occlusal loads in such a manner as to reduce the chance of tooth fracture.

Dental Rehabilitation with Cast Metal Alloys

When cast metal restorations have been *used to restore adjacent or opposing* teeth, the continued use of the same material may be considered to eliminate electrical and corrosive activity that sometimes occurs between dissimilar metals in the mouth, particularly when they come in *contact* with each other.

Posterior Diastema Closure and Occlusal Plane Correction

I. The cast inlay or onlay is indicated when extension of the mesiodistal dimension of the tooth is necessary to form a *contact with an adjacent tooth*.
II. Cast onlays also can be used to *correct the occlusal plane* of a slightly tilted tooth.

Removable Prosthodontic Abutment

Teeth that are to serve as abutments for a removable partial denture can be restored with a cast metal restoration. The major advantages of a cast restoration are as follows:

I. The superior physical properties of the cast metal alloy allow it to better withstand the forces imparted by the partial denture.
II. The rest seats, guiding planes and other aspects of contour relating to the partial denture are better controlled when the indirect technique is used.

Endodontically Treated Teeth

A molar or premolar with treatment root canal filling can be restored with a cast metal onlay, provided that the onlay has been thoughtfully designed to distribute occlusal loads in such a manner as to reduce the chance of tooth fracture.

Contraindications

High Caries Rate

Facial and lingual (especially lingual) smooth-surface caries indicates a *high caries activity* that should be brought under control before expensive cast metal restorations are used.

Young Patients

With younger patients, *direct restorative materials* (e.g. composite or amalgam) are indicated, unless the tooth is severely broken or endodontically treated. An indirect procedure in a young patient has the following disadvantages:

I. Treatment requires longer and more numerous appointments
II. Access is more difficult
III. Clinical crowns are shorter with larger pulp chambers
IV. Younger patients may neglect oral hygiene, resulting in additional caries.

Esthetics

The dentist must consider the esthetic impact (display of metal) of the cast metal restoration. This factor usually limits the use of cast metal restorations to tooth surfaces that are invisible at a conversational distance. Composite and porcelain restorations are alternatives in esthetically sensitive areas.

Small Restorations

Amalgam and composites are the materials of choice for smaller restorations. Cast metal inlays are rarely done in small class I and II restorations.

Advantages

Strength

The inherent strength of dental casting alloys allows them to restore large damaged or missing areas and be used in ways that protect the tooth from future fracture injury.

Biocompatibility

As previously mentioned, *high-gold dental casting alloys* are unreactive in the oral environment. This biocompatibility can be helpful for many patients who have allergies or sensitivities to other restorative materials.

Low Wear

Although individual casting alloys vary in their wear resistance, castings are able to withstand occlusal loads with minimal changes. This is especially important in *large restorations* that restore a large percentage of occlusal contacts.

Control of Contours and Contacts

Through the use of the indirect technique, the dentist has great control over contours and contacts. This control becomes especially important when the restoration is larger and more complex.

Disadvantages

Number of Appointments and Higher Chair Time

The cast inlay or onlay requires at least *two appointments* and much more time than a direct restoration, such as amalgam or composite.

Temporary Restorations

Patients must have temporary restorations between the preparation and delivery appointments. Temporaries occasionally loosen or break, requiring additional visits.

Cost

In some instances, cost to the patient becomes a major consideration in the decision to restore teeth with cast metal restorations. The cost of materials, laboratory bills and the time involved make indirect cast restorations more expensive than direct restorations.

Technique Sensitivity

Every step of the indirect procedure requires diligence and attention to detail. Errors at any part of the long, multistep process tend to be compounded, resulting in less than ideal fits.

Splitting Forces

Small inlays may produce a wedging effect on facial or lingual tooth structure and increase the potential for splitting the tooth. *Onlays do not have this disadvantage.*

Preoperative Procedures

Occlusion

The occlusal contacts of teeth should be evaluated before the administration of the anesthetic or tooth preparation. An evaluation should include:

I. The occlusal contacts in *maximum inter-cuspation* where teeth are brought into full interdigitation.
II. The occlusal contacts that occur during *mandibular movements*.

Anaesthesia

Local anaesthesia of the tooth to be operated on and of adjacent soft tissue usually is recommended. Anaesthesia in these areas eliminates pain and reduces salivation, resulting in a more pleasant procedure for the patient and the operator.

Considerations for Temporary Restorations

Before tooth preparation, consideration must be given to the method that will be used to fabricate the temporary restoration. Most temporary restoration techniques require the use of a *preoperative impression* to reproduce the occlusal, facial and lingual surfaces of the temporary restoration to the preoperative contours.

I. *Alginate impression* materials may be used and are relatively inexpensive.
II. Polyvinyl siloxane (PVS) impression material may be used if additional accuracy, stability and durability are required.

Tooth Preparation for Class II Cast Metal Inlay

The following section would give a step-wise illustration of the preparation of a class II disto-occlusal (DO) cast metal inlay in a maxillary second premolar (Fig. 23.1A):

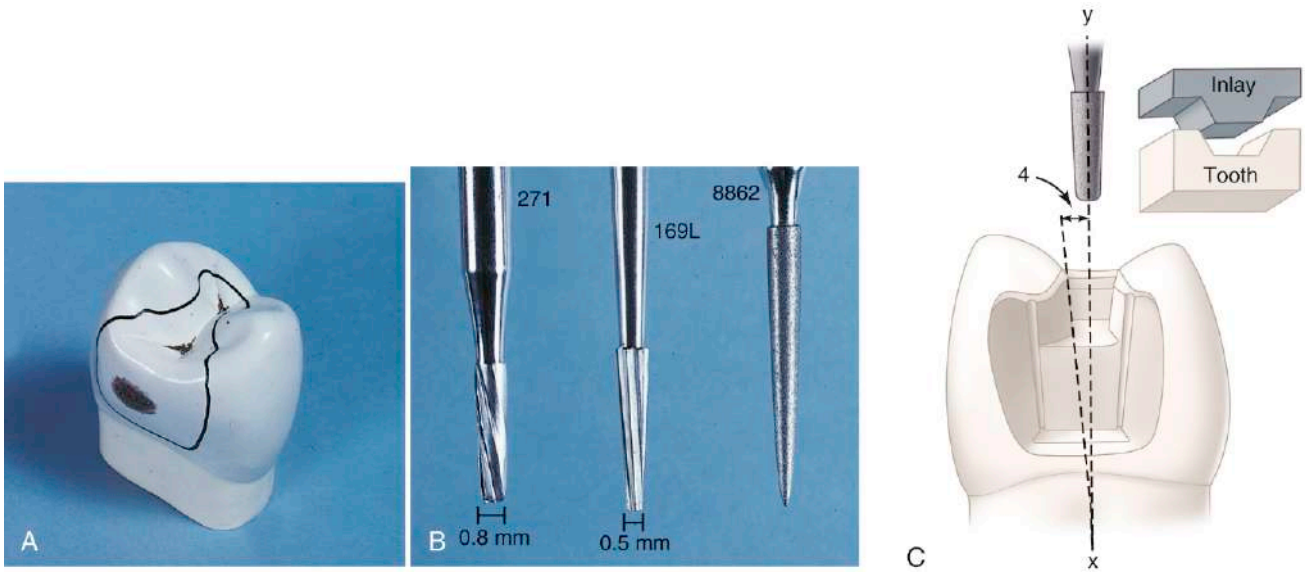


Figure 23.1 (A) Proposed outline form for disto-occlusal preparation in a maxillary premolar. (B) Dimensions and configuration of No. 271, No. 169L and No. 8862 instruments. (C) Conventional 4 degree divergence from line of draw (line *xy*).

Armamentarium

I. *Plane cut, tapered fissure carbide burs* are used to develop the vertical internal walls of the preparation for cast metal inlays and onlays. The recommended burs are:
   i. **No. 271 bur** (Brasseler USA, Inc).
   ii. **No. 169L bur** (Brasseler USA, Inc).

II. A *slender, fine-grit, flame-shaped diamond instrument* is used to place the marginal bevels. The recommended diamond is:
   i. **No. 8862 bur** (Brasseler USA, Inc).

- The sides and end surface of the No. 271 bur meet in a *slightly rounded manner* so that sharp, stress-inducing internal angles are not formed in the preparation.[4]
- The burs used to develop the vertical walls are oriented to a *single 'draw' path*, usually the long axis of the tooth crown, so that the completed preparation has no undercuts (Fig. 23.1C).
- The gingival-to-occlusal divergence of these preparation walls may range from *2 to 5 degrees per wall* from the line of draw. If the vertical walls are unusually short, a maximum of 2 degree occlusal divergence is desirable to increase retention potential. As the occlusogingival height increases, the occlusal divergence should increase because lengthy preparations with minimal divergence (more parallel) may present difficulties during the seating and withdrawal of the restoration.
- Recommended dimensions and configurations of the burs to be used are shown in Fig. 23.1B.

The tooth preparation for a class II inlay is summarized in Table 23.1.

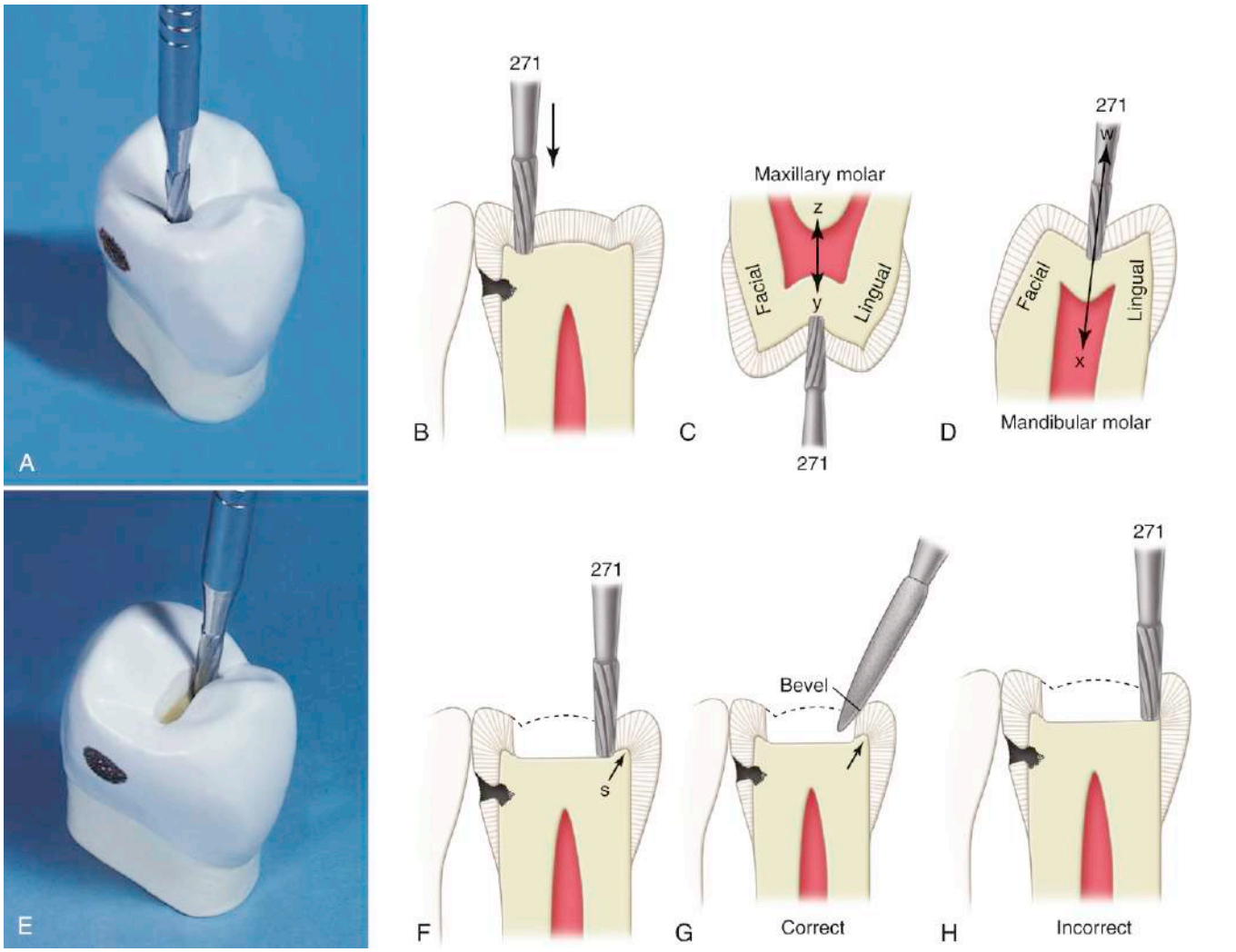


Figure 23.2 (A and B) Bur after punch cut to a depth of 1.5 mm. (C) For maxillary posterior teeth, the long axis of the bur should parallel the long axis of the tooth crown (line *yz*). (D) For molar and second premolar teeth of mandibular dentition, the long axis of the bur should tilt slightly lingually to parallel the long axis of the tooth crown (line *wx*). (E and F) Extending the mesial wall, taking care to conserve dentin that supports marginal ridge (*s*). (G) The marginal bevel can provide additional extension. (H) Improper extension that has weakened the marginal ridge.

Table 23.1 Clinical steps in tooth preparation of a class II inlay

I. Initial preparation
  i. Occlusal steps
    – Step 1: Orienting the bur
    – Step 2: Occlusal punch cut
    – Step 3: Occlusal extension
    – Step 4: Dovetail retention
    – Step 5: Occlusal outline form
  ii. Proximal box
    – Step 1: Proximal ditch preparation
    – Step 2: Proximal box preparation
    – Step 3: Planing the walls
    – Step 4: Placement of retention grooves

II. Final preparation
  i. Removal of infected carious dentin and pulp protection
    – Step 1: Inspection
    – Step 2: Removal of infected caries
    – Step 3: Removal of old restorative material
    – Step 4: Pulp protection with light cure GIC
    – Step 5: Lining with calcium hydroxide (if required)
  ii. Preparation of bevels and flares
    – Step 1: Preparation of occlusal bevel
    – Step 2: Bevelling the axiopulpal line angle
    – Step 3: Preparing the secondary lingual flare
    – Step 4: Bevelling the gingival margin
    – Step 5: Preparing the secondary facial flare

Initial Preparation

Occlusal Step

Step 1: Orienting the bur

- The *No. 271 carbide bur* is held parallel to the long axis of the tooth crown.
- The bur should be rotating at high speed (with air–water spray) before application to the tooth and should not stop rotating until it is removed; this minimizes perceptible vibration and prevents breakage or chipping of the bur blades.
- A general rule is to maintain the long axis of the bur parallel to the long axis of the tooth crown at all times (Fig. 23.2B and C).
- For mandibular molars and second premolars whose crowns tilt slightly lingually, this rule dictates that the bur should also be tilted slightly (5–10 degrees) lingually to conserve the strength of the lingual cusps (Fig. 23.2D).

Step 2: Occlusal punch cut

Enter the fossa or pit closest to the involved marginal ridge, using a *punch cut* to a depth of *1.5 mm* to establish the depth of the pulpal wall (Fig. 23.2A and B). In the initial preparation, this specified depth should not be exceeded, regardless of whether the bur end is in dentin, caries, old restorative material or air.

Step 3: Occlusal extension

Maintaining the 1.5 mm initial depth and the same bur orientation, the dentist extends the preparation outline along the central groove or fissure to include the mesial fossa or pit (Fig. 23.2E and F). Ideally, the *faciolingual dimension of this cut should be minimal.*

The dentist takes care to keep the mesial marginal ridge strong by not removing the dentin support of the ridge (Fig. 23.2F and H).

Occasionally, a fissure extends onto the mesial marginal ridge. This defect, if shallow, may be treated with *enamoloplasty*, or it may be included in the outline form with the cavosurface bevel, which is applied in a later step in the tooth preparation (Fig. 23.2G).

Step 4: Dovetail retention

I. Use the slender **169L** carbide bur to create the facial and lingual extensions into the mesial pit region, thereby creating the desired *dovetail retention form*. This resists distal displacement of the inlay (Fig. 23.3B).

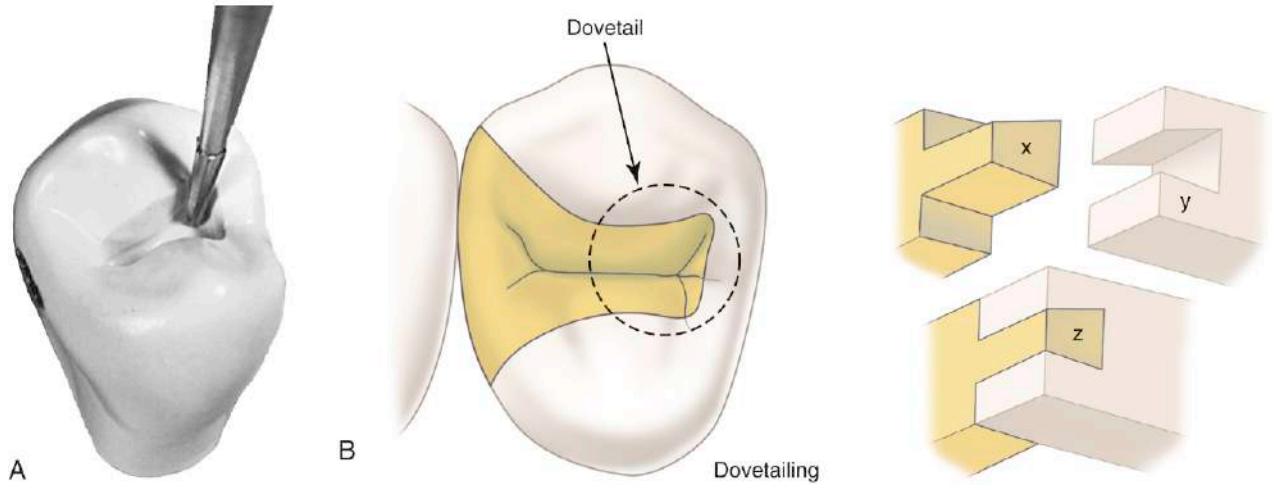


Figure 23.3 (A) Extending up the mesiofacial triangular groove using the slender No. 169L bur. (B) Dovetail retention form is created by extension shown in *A*. As *x* fits into *y* only in one direction resulting in *z*, similarly dovetail portion of inlay fits into the dovetail portion of the preparation only in an occlusal-to-gingival direction.

II. Extend to include *faulty facial and lingual fissures radiating from the mesial pit*. During this extension cutting, the operator is cautioned again not to remove the dentin support of the proximal marginal ridge (Fig. 23.3A).

Step 5: Occlusal outline form

Use the 271 carbide bur to extend the occlusal step distally into the distal marginal ridge (Fig. 23.4A and B). While extending distally, progressively widen the preparation to the desired faciolingual width in anticipation of the proximal box preparation.

The increased faciolingual width enables the facial and lingual walls of the box to project (visually) perpendicularly to the proximal surface at positions that *clear the adjacent tooth by 0.2–0.5 mm* (Fig. 23.4F).

The facial and lingual walls of the occlusal step should go around the cusps in *graceful curves*, and the prepared isthmus in the transverse ridge ideally should be only

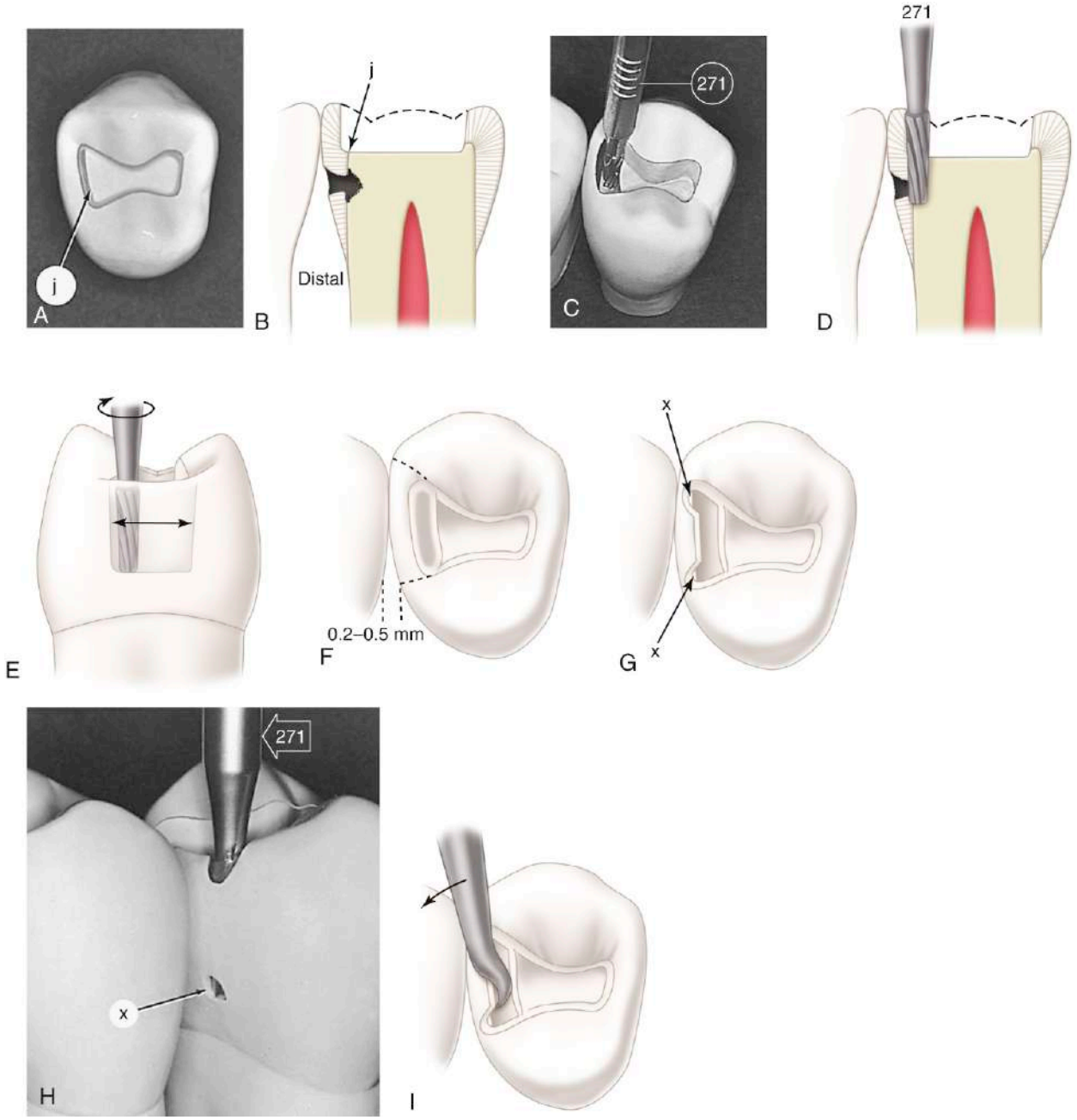


Figure 23.4 (A) After exposing the junction (*j*) of proximal enamel and dentin. (B) Sectional drawing of *A*. (C) Cutting the proximal ditch. (D) Sectional drawing of *C*. (E) Proximal view of *D*. (F) Occlusal view of the proximal ditch with proposed ideal clearance with the adjacent tooth. (G and H) Proximal ditch extended distally. *x*, penetration of enamel by side of bur at its gingival end. (I) Breaking away isolated enamel.

slightly wider than the bur, thus conserving the dentinal protection for the pulp and maintaining the strength of the cusps.

Proximal box

Step 1: Proximal ditch preparation

- Continuing with the No. 271 carbide bur, the distal enamel is isolated by cutting a *proximal ditch* (Fig. 23.4C–F). The mesiodistal width of the ditch should be 0.8 mm (the tip diameter of the bur) and prepared approximately *two-thirds* (*0.5 mm*) *at the expense of dentin and one-third* (*0.3 mm*) *at the expense of enamel.*
- The gingival extension of this cut may be checked with the length of the bur by first measuring the depth from the height of the marginal ridge and then removing the bur and holding it beside the tooth. A *periodontal probe* also may be used for this measurement.
- The harder enamel should *guide* the bur. While penetrating gingivally, the dentist extends the proximal ditch facially and lingually *beyond the caries* to the desired position of the facioaxial and linguoaxial line angles.
- Ideal gingival extension provides a *0.5-mm clearance of the unbevelled gingival margin* with the adjacent tooth.
- Moderate to extensive caries on the proximal surface dictates continued extension of the proximal ditch to the *extent of the caries* at the dentinoenamel junction (DEJ), but not pulpally.
- The operator should *guard against overcutting* the facial, lingual and gingival walls, which would not conserve the tooth structure and could result in:
  - overextension of the margins in the completed preparation,
  - a weakened tooth and
  - possible injury of soft tissue.
- As the proximal enamel diminishes in thickness from the occlusal to gingival level, the end of the *bur is closer to the external tooth surface* as the cutting progresses gingivally.

Step 2: Proximal box preparation

- With the No. 271 carbide bur, make *two cuts*, one at the facial limit of the proximal ditch and the other at the lingual limit (Fig. 23.4G). These cuts are extended until the bur is *nearly through the marginal ridge enamel* (the side of the bur may emerge slightly through the surface at the level of the gingival floor) as shown in Fig. 23.4H.
- This step weakens the enamel by which the remaining isolated portion is held.
- The level of the gingival floor is verified by observing where the end of the bur emerges through the proximal surface.
- If indicated, additional gingival extension can be accomplished while the remaining enamel still serves to guide the bur and to prevent it from marring the proximal surface of the adjacent tooth. The isolated wall of enamel if still present, can be fractured out with a spoon excavator (Fig. 23.4I).

Step 3: Planing the walls

- *Planing* the distofacial, distolingual and gingival walls is accomplished by hand instruments in order to remove all undermined enamel. Depending on access, the operator can use a bin-angle chisel (Fig. 23.5) or *enamel hatchet*.
- The dentist planes the wall by holding the instrument in the modified palm-and-thumb grasp and uses a chisel-like motion in an occlusal-to-gingival direction (Fig. 23.5A and B).
- The dentist planes the gingival wall by using the same instrument as a hoe, scraping in a lingual-to-facial direction (Fig. 23.5C).
- When proximal caries is minimal, ideal facial and lingual extensions at this step in the preparation result in margins that clear the adjacent tooth by *0.2–0.5 mm*.

Step 4: Placement of retention grooves

- Shallow (0.3-mm deep) *retention grooves* may be cut in the facioaxial and linguoaxial line angles with the **No. 169L** carbide bur (Fig. 23.5E–I).
- These grooves are indicated especially when the prepared tooth is short.
- When properly positioned, the grooves are in sound dentin, close to but not contacting, the DEJ.
- The long axis of the bur must be held parallel to the line of draw.

Final Preparation

Removal of Infected Carious Dentin and Pulp Protection

Step 1: Inspection

After the initial preparation has been completed, the dentist evaluates the internal walls of the preparation visually and tactilely (with an explorer) for indications of any remaining carious dentin.

If carious dentin remains, and if it is judged to be infected, but shallow or moderate (≥1 mm of remaining dentin between the caries and the pulp), satisfactory isolation for the removal of such caries is done with the use of cotton rolls, a saliva ejector and gingival retraction cord.

The retraction cord also serves to widen the gingival sulcus and slightly retract the gingiva in preparation for bevelling and flaring the proximal margins (Fig. 23.6).

Step 2: Removal of infected caries

A slowly revolving *round bur* (No. 2 or No. 4) or *spoon excavator* is used to remove carious infected dentin (Fig. 23.6F and G).

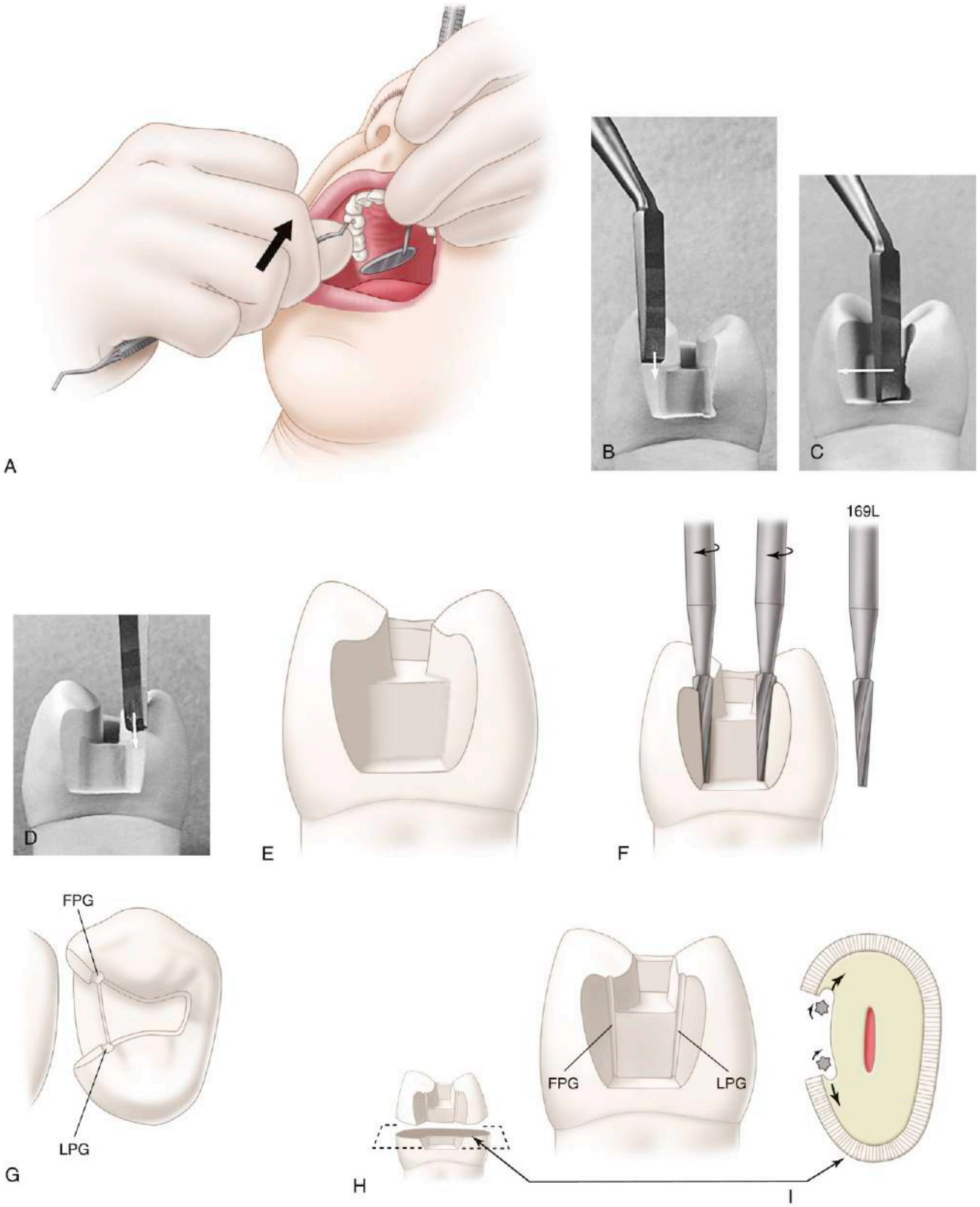


Figure 23.5 (A–D) Using modified palm-and-thumb grasp (*A*) to plane distofacial and distolingual walls (*B* and *D*) and to scrape gingival wall (*C*). (E) Before cutting retention grooves. (F) Cutting retention grooves. (G and H) Facial proximal groove (*FPG*) and lingual proximal groove (*LPG*). (I) Section in plane *x*. *Large arrows* depict the direction of translation of the rotating bur.

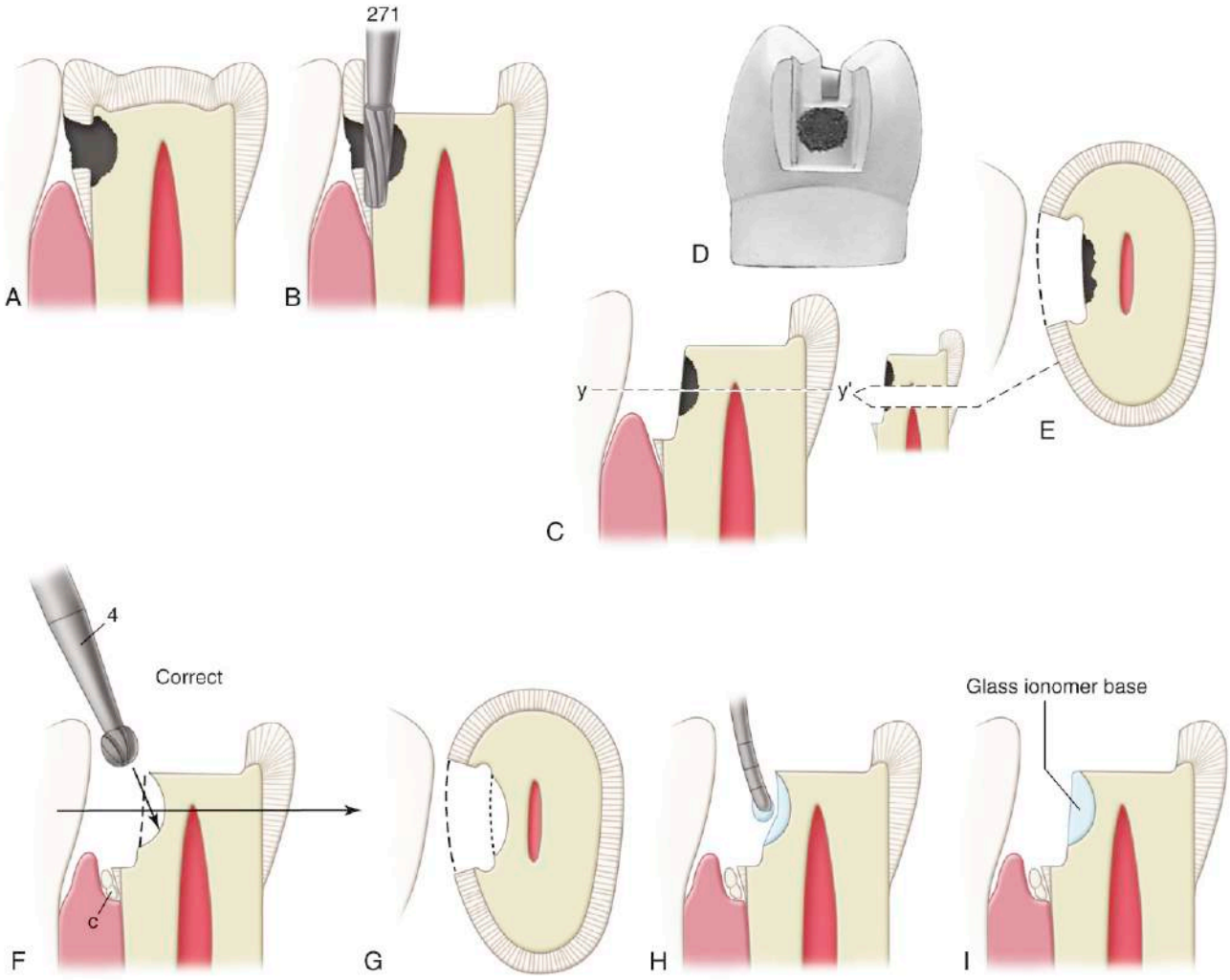


Figure 23.6 Moderately deep caries. (A–C) Extending the proximal ditch gingivally (*B*) to a sound floor free from caries (*C*). (D) Remaining caries on the axial wall. (E) Section of *C* in plane *yy′*. (F) Removing the remaining infected dentin. *c*, inserted retraction cord. (G) Section of *F*. (H) Inserting resin modified glass ionomer base with periodontal probe. (I) Completed base.

This excavation is done just above stall-out speed with light, intermittent cutting.

The operator should avoid unnecessarily desiccating the exposed dentin during this procedure.

Step 3: Removal of old restorative material

Any remaining old restorative material on the internal walls should be removed if any of the following conditions are present:

I. The old material is judged to be thin, nonretentive or both.
II. Radiographic evidence of caries under the old material is present.
III. The pulp was symptomatic preoperatively.
IV. The periphery of the remaining restorative material is not intact (i.e. some breach exists in the junction of the material with the adjacent tooth structure that may indicate caries under the material).

If none of these conditions is present, the operator may elect to leave the remaining restorative material to serve as a base, rather than risk unnecessary removal of sound dentin or irritation or exposure of the pulp.

Step 4: Pulp protection with light cure GIC

Light-cured glass ionomer cement may be mixed and applied with a suitable applicator to these excavated regions to the depth and form of the ideally prepared surface.

- Placing a *base takes little time* and should be considered because it results in working dies that have preparation walls with no undercuts and 'ideal' position and contour.
- Applying a base at this time *minimizes additional irritation of the pulp* during subsequent procedures necessary for the completion of the restoration.

- The light-cured glass ionomer *adheres to the tooth structure* and does not require retentive undercuts when the base is small to moderate.
- The material is applied by conveying *small portions* on the end of a periodontal probe and is light-cured when the correct form has been achieved (Fig. 23.6H and I).
- Any excess cement can be trimmed back to the ideal form with the No. 271 carbide bur after the cement has hardened.
- When excavating extensive caries, the dentist attempts to remove *only* infected dentin and not affected dentin because removal of the latter might expose a healthy pulp.
- If removal of soft, infected dentin leads directly to a pulpal exposure (carious pulpal exposure), root canal treatment should be accomplished before completing the cast metal restoration.
- If the pulp is inadvertently exposed as a result of operator error or misjudgment (mechanical pulpal exposure), the operator must decide whether to proceed with the root canal treatment or to attempt a *direct pulp capping procedure*.
- A *favourable prognosis* for the pulp after direct pulp capping may be expected if the following criteria are met:
  - The exposure is *small* (<0.5 mm in diameter).
  - The tooth has been asymptomatic, showing no signs of pulpitis.
  - Any haemorrhage from the exposure site is easily controlled.
  - The invasion of the pulp chamber was relatively atraumatic with little physical irritation to the pulp tissue.
  - A clean, uncontaminated operating field is maintained (i.e. by using a rubber dam).

Step 5: Lining with calcium hydroxide (if required)

If the excavation closely approaches the pulp or if a direct pulp cap is indicated, the dentist should first apply a lining of calcium hydroxide using a *flow technique* (without pressure). This calcium hydroxide liner should cover and protect any possible near or actual exposure and extend over a major portion of the excavated dentinal surface (Fig. 23.7A).

- Calcium hydroxide treatment of an exposed, healthy pulp promotes the formation of a *dentin bridge*, which would close the exposure.[3]
- The peripheral 0.5–1 mm of the dentin excavation should be left available for bonding the subsequently applied light-cured glass ionomer cement base.
- If a calcium hydroxide liner has been applied, less dentin is available for adhesive bonding. In these circumstances, small mechanical undercuts can increase the retention of the glass ionomer base. *Retention coves* are placed with the No. ¼ carbide bur (Figs. 23.7B–D and 23.8). These coves are placed in the peripheral dentin of the excavation and are as remote from the pulp as possible.

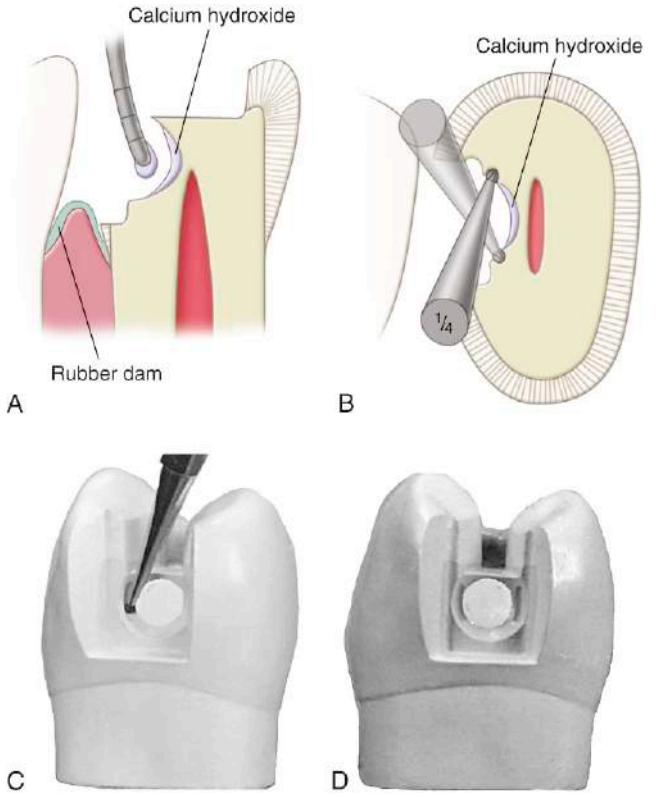


Figure 23.7 (A) Deep caries excavations near the pulp are first lined with calcium hydroxide. Note the rubber dam. (B–D) Cutting retention coves for retaining glass ionomer cement.

Preparation of Bevels and Flares

Concept

I. *Bevelling* the occlusal and gingival margins and applying the *secondary flare* on the distolingual and distofacial walls results in a *30–40 degree marginal metal on the inlay* (Figs. 23.9B, 23.10J and 23.11F).
II. This cavosurface design helps seal and protects the margins and results in a strong enamel margin with an angle of 140–150 degrees.

Armamentarium

I. The slender, flame-shaped, fine-grit diamond instrument ***No. 8862 bur*** (Brasseler USA, Inc) is used for both bevelling the margins and for preparing the secondary flare.
II. The slender ***No. 169L carbide bur*** can also be used for preparing the secondary flare.

Step 1: Preparation of occlusal bevel

- The width of the cavosurface bevel on the occlusal margin should be approximately *one-fourth the depth of the respective wall* (Fig. 23.9A and B).

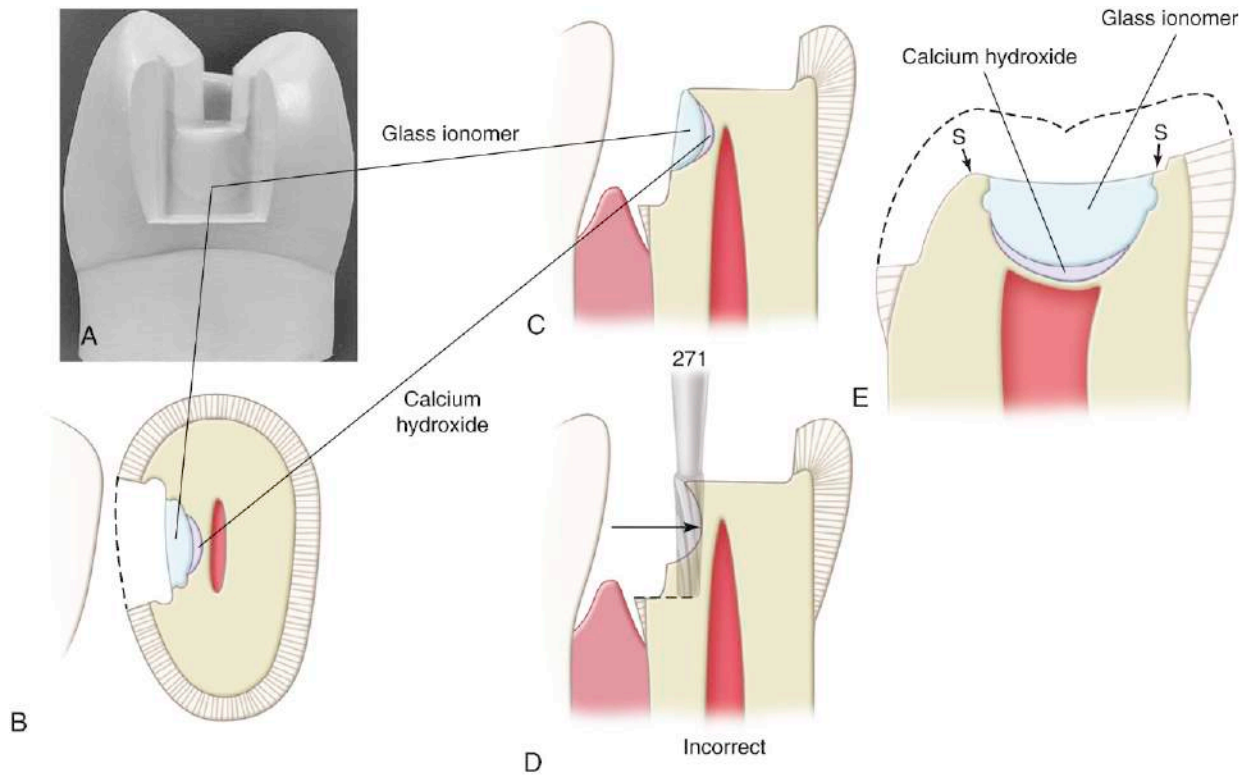


Figure 23.8 (A–C) Completed base for the treatment of deep caries. (D) Never deepen entire axial wall with the side of a fissure bur to remove caries because the pulp would be greatly irritated from the resulting closeness of the gingivoaxial region of the preparation. (E) Cement base placed deep in the excavation on the mandibular molar. Note the flat seats in sound dentin (*S*) that are required for adequate resistance form.

- The resulting occlusal marginal metal of the inlay should be *40 degree metal*; the occlusal marginal enamel is *140 degree enamel* (Fig. 23.9B and E). Bevelling the occlusal margins in this manner increases the strength of the marginal enamel and helps seal the margins.
- While bevelling the occlusal margins, a guide to diamond positioning is to maintain an approximate *40 degree angle* between the side of the instrument and the external enamel surface.
- The margins on the proximal marginal ridges always require a cavosurface bevel (Fig. 23.9D and I). Failure to apply a bevel in these regions leaves the enamel margin weak and subject to injury by fracture.

Step 2: Bevelling the axiopulpal line angle

The diamond instrument is used to bevel the axiopulpal line angle lightly (Fig. 23.9D). Such a bevel provides a thicker and stronger wax pattern at this critical region. The desirable *metal angle at the margins of inlays is always 40 degrees except at the gingival margins, where the metal angle should be 30 degrees*.

Step 3: Preparing the secondary lingual flare

- The distolingual wall should extend from the linguoaxial line angle into the lingual embrasure in two planes (Fig. 23.10A). The first flare that brings the proximal preparation out of the contact area is termed *lingual primary flare*; the second flare that is given to provide a marginal metal angle of 40 degrees is termed *lingual secondary flare*.
- Using the flame-shaped diamond instrument that is rotating at high speed, the dentist prepares the lingual secondary flare (Figs. 23.11B–D and 23.10A). The dentist approaches from the lingual embrasure (Fig. 23.11D), moving the instrument mesiofacially.

A cavosurface enamel angle of *more than 150 degrees is incorrect* because it results in a less defined enamel margin (finish line), and the marginal cast metal alloy is too thin and weak if its angle is less than 30 degrees.

- Conversely, if the enamel margin is *140 degrees or less*, the metal is too bulky and difficult to burnish when its angle is greater than 40 degrees (Fig. 23.9F).
- Usually, it is helpful to insert a gingival retraction cord of suitable diameter into the gingival sulcus adjacent to the gingival margin before the use of the flame-shaped diamond instrument on the proximal margins (Fig. 23.11A).

Step 4: Bevelling the gingival margin

- The dentist bevels the gingival margin by moving the instrument facially along the gingival margin (Figs. 23.10A and 23.11E). The instrument should be *tilted slightly mesially* to produce a gingival bevel with

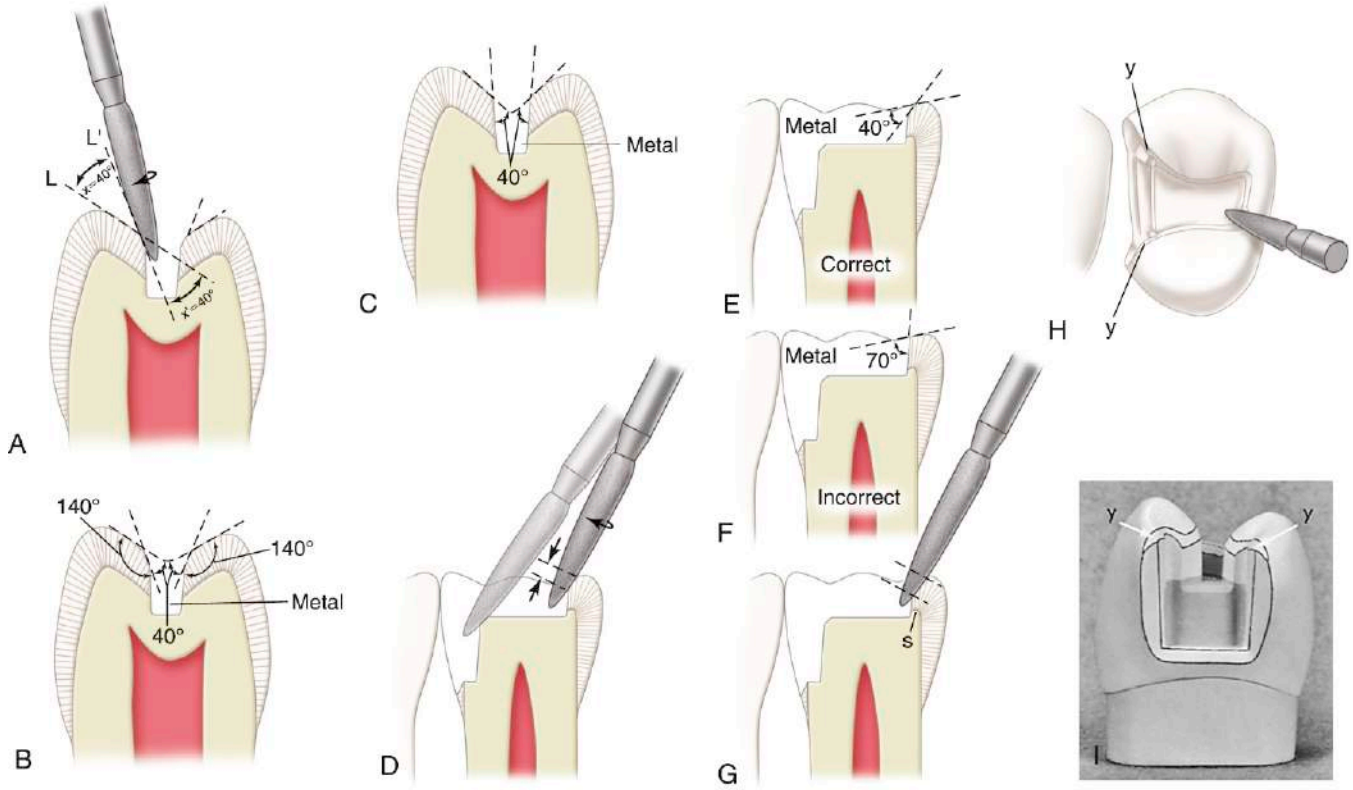


Figure 23.9 (A) The diamond instrument bevelling the occlusal margin when it is indicated to result in 40 degree marginal metal, as shown in *B*. Angles *x* and *x′* are equal because the opposite angles are equal when two lines (*L* and *L′*) intersect. The diamond instrument is always directed such that an angle of 40 degrees is made by the side of the instrument and the external enamel surface. (B) Occlusal marginal metal is approximately 40 degrees in cross-section, making the enamel angle 140 degrees. (C) When the cuspal inclines are steep, no bevelling is indicated considering that 40 degree metal would result without bevelling. (D) Bevelling the mesial margin and the axiopulpal line angle. (E) The mesial bevel is directed correctly to result in 40 degree marginal metal. (F) An unbevelled mesial margin is incorrect because it results in a weak enamel margin and unburnishable marginal metal. (G) To conserve dentinal support (*s*), occlusal defects on the marginal ridge are included in the outline form by applying a cavosurface bevel, which may be wider than usual, when necessary. (H) Occlusal view of *G*. Preparing a 140 degree cavosurface enamel angle at regions labelled *y* usually dictates that the occlusal bevel be extended over the marginal ridges into the secondary flares. (I) Distal view of *H*.

the correct steepness to result in 30 degree marginal metal (Fig. 23.11G and H).

- If the instrument is not tilted in this manner, the bevel is too steep, resulting in gingival bevel metal that is too thin (<30 degree metal) and too weak.
- Although the instrument is tilted mesially, its long axis must not tilt facially or lingually (Fig. 23.11E).
- The gingival bevel should be *0.5–1 mm wide* and should blend with the lingual secondary flare.
- The gingival bevel serves the following purposes:
  - *Weak enamel is removed* (Fig. 23.11J).
  - The bevel results in 30 degree metal that is *burnishable* (on the die) because of its angular design (Fig. 23.11F).
  - A *lap, sliding fit* is produced at the gingival margin (Fig. 23.11G and H).

Step 5: Preparing the secondary facial flare

- The long axis of the instrument during this secondary facial flare (Fig. 23.10A–F) is again returned nearly to the line of draw, with only a small tilting mesially and facially, and the direction of translation of the instrument is that which results in *40 degree marginal metal* (Fig. 23.10E and J).
- Care must be exercised to avoid abrading the adjacent tooth and overextending the distofacial margin. To prevent such abrasion or overextension, the instrument may be *raised occlusally* (using the narrower portion at its tip end) to complete the most facial portion of the wall and margin (Fig. 23.10D).
- The more slender *No. 169L carbide bur* may be used, rather than the flame-shaped diamond instrument (Fig. 23.10H). The No. 169L bur produces an extremely smooth surface to the secondary flare and a smooth, straight distofacial margin.
- The *secondary flare* is necessary for several reasons:
  - The secondary flaring of the proximal walls extends the margins into the embrasures, making these margins more *self-cleaning* and more *accessible to finishing procedures* during the inlay insertion appointment.

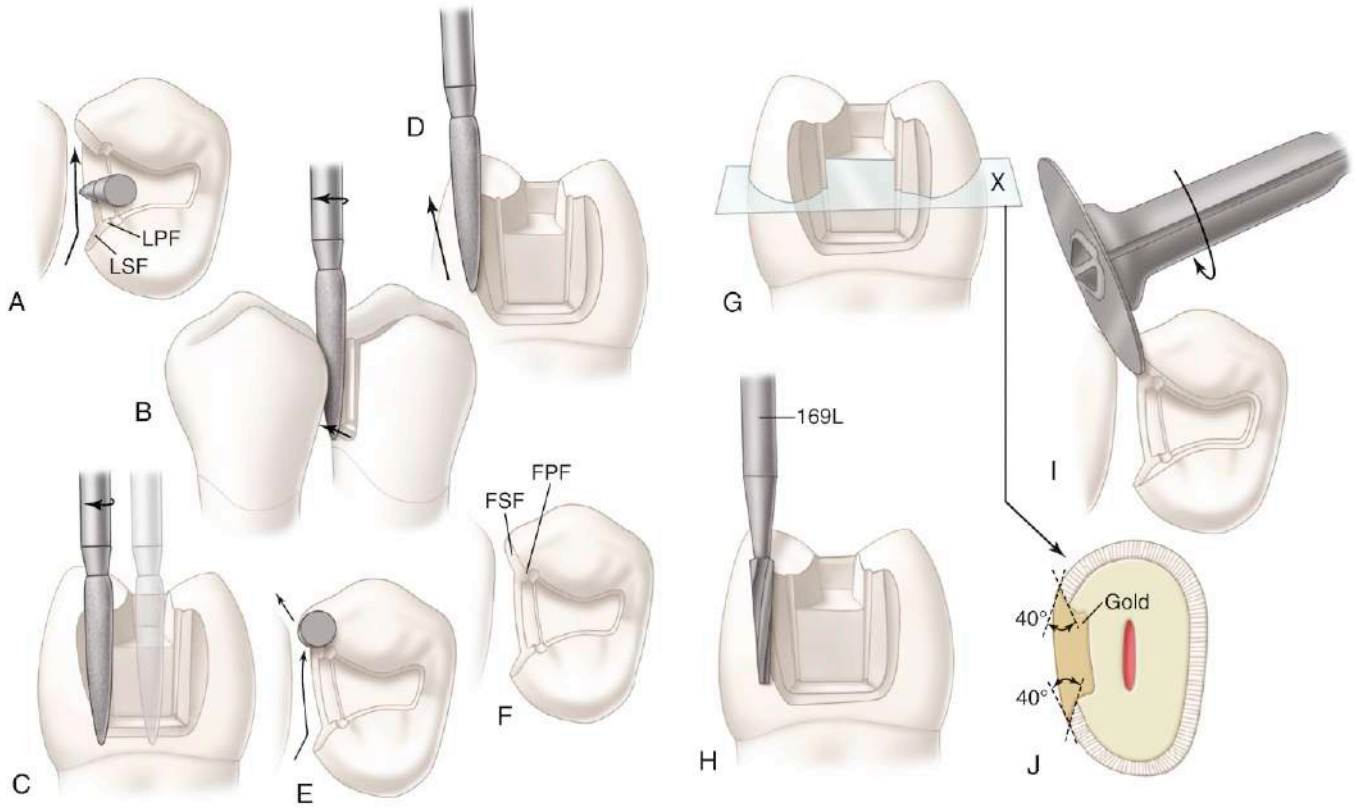


Figure 23.10 (A) Occlusal view of Fig. 23.11E. *LSF*, lingual secondary flare; *LPF*, lingual primary flare. (B–E) Preparing the facial secondary flare. *Large arrows* in *B*, *D* and *E* indicate the direction of the translation. (F) Completed facial secondary flare. *FSF*, facial secondary flare; *FPF*, facial primary flare. (G) Distal view of *F*. *x*, Plane of cross-section shown in *J*. (H and I) Preparing the secondary flare with the No. 169L carbide bur (*H*) or with paper disk (*I*). (J) The secondary flares are directed to result in 40 degree marginal metal and 140 degree marginal enamel.

- The direction of the flare results in *40 degree marginal metal* (Fig. 23.10J). Metal with this angular design is burnishable.
- A more blunted and *stronger enamel margin* is produced because of the secondary flare.

The completed preparation is illustrated in Fig. 23.12.

Modifications in Inlay Tooth Preparations

Because the indications for small inlays are rare, the following sections provide procedural information that may promote better understanding of their applications in more complex and larger inlay or onlay restorations.

Mesio-occluso-distal Preparation

If a marginal ridge is severely weakened because of excessive extension, the preparation outline often should be altered to include the proximal surface. The DO preparation illustrated in the previous section would be extended to a mesio-occluso-distal (MOD) preparation (Figs. 23.12B–D and 23.13). The decision to extend the preparation in this manner calls for clinical judgment as to whether the remaining marginal ridge would withstand occlusal forces without fracture. A fortunate factor in favour of not extending the preparation is that such ridge enamel usually is composed of gnarled enamel and is stronger than it appears. Caries present on both proximal surfaces would result in a MOD preparation and restoration. The only difference in technique as described previously is the inclusion of the other proximal surface.

Modifications of Class II Preparation for Esthetics

For esthetic reasons, minimal flare is desired for the mesiofacial proximal wall in maxillary premolars and first molars in class II cast metal preparations (Fig. 23.12D). The mesiofacial margin is minimally extended facially of the contact to such a position that the margin is barely visible from a facial viewing position.

To accomplish this, the secondary flare is omitted, and the wall and margin are developed with:

I. a chisel or enamel hatchet and final smoothing with a fine-grit paper disk or
II. a narrow diamond or bur when access permits.

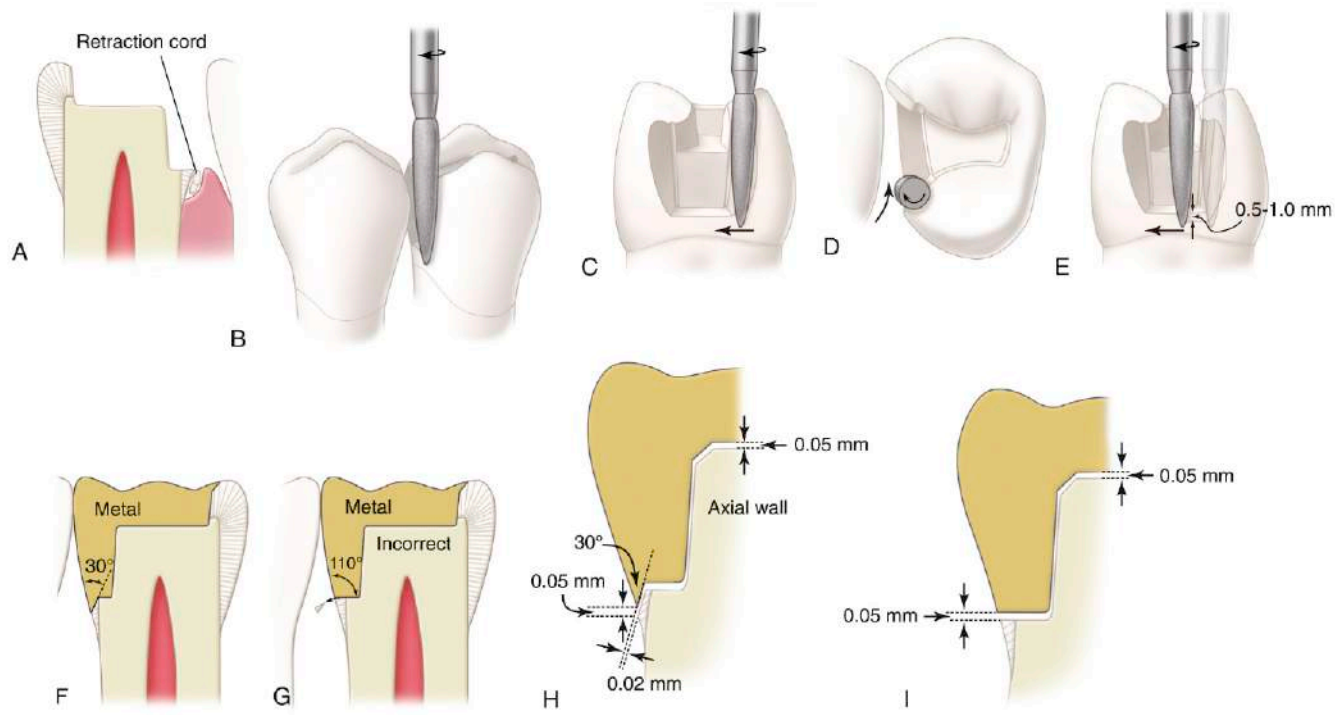


Figure 23.11 (A) The retraction cord is inserted in the gingival sulcus and left for several minutes. (B–D) Diamond instrument preparing lingual secondary flare. Large arrow in *D* indicates the direction of the translation. (E) Bevelling the gingival margin. (F) Properly directed gingival bevel resulting in 30 degree marginal metal. (G) Failure to bevel the gingival margin results in a weak margin formed by undermined rods (note the easily displaced wedge of enamel) and 110 degree marginal metal, an angular design unsuitable for burnishing. (H) Lap, sliding fit of prescribed bevel metal decreases the 50 μm error of seating to 20 μm. (I) A 50-μm error of seating produces an equal cement line of 50 μm along the unbevelled gingival margin.

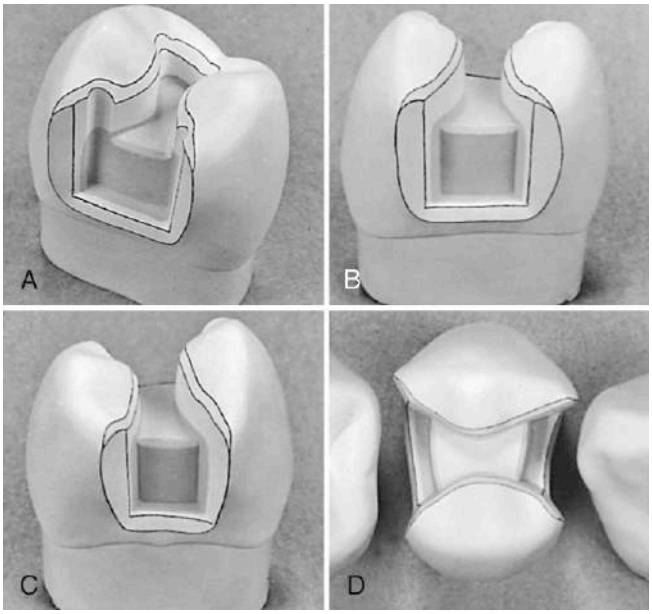


Figure 23.12 (A) Completed disto-occlusal preparation for the inlay. (B) Mesio-occluso-distal preparation for the inlay on the maxillary right first premolar, disto-occlusal view. (C) Same preparation as in B, mesio-occlusal view. (D) Same preparation as in B, occlusal view. Note the absence, for esthetic reasons, of secondary flare on the mesiofacial aspect and minimal extension of the mesiofacial margin.

Facial or Lingual Surface Groove Extension

Sometimes, a faulty facial groove (fissure) on the occlusal surface is continuous with a faulty facial surface groove (mandibular molars), or a faulty distal oblique groove on the occlusal surface is continuous with a faulty lingual surface groove (maxillary molar). This situation requires extension of the preparation outline to include the fissure to its termination (Fig. 23.14). Occasionally the operator may extend further gingivally than the fissure length to improve retention form. Such groove extensions, when sufficiently long, are effective for increasing retention. Likewise, this extension may be indicated to provide sufficient retention form even though the facial or lingual surface grooves are not fissured.

For extension onto the facial surface, the dentist uses the No. 271 carbide bur held parallel to the line of draw and extends through the facial ridge (Fig. 23.14A and B). The depth of the cut should be 1.5 mm. The floor (pulpal wall) should be continuous with the pulpal wall of the occlusal portion of the preparation (Fig. 23.14D).

With the bur still aligned with the path of draw, the dentist uses the side of the bur to cut the facial surface portion of this extension (Fig. 23.14C). The diameter of the

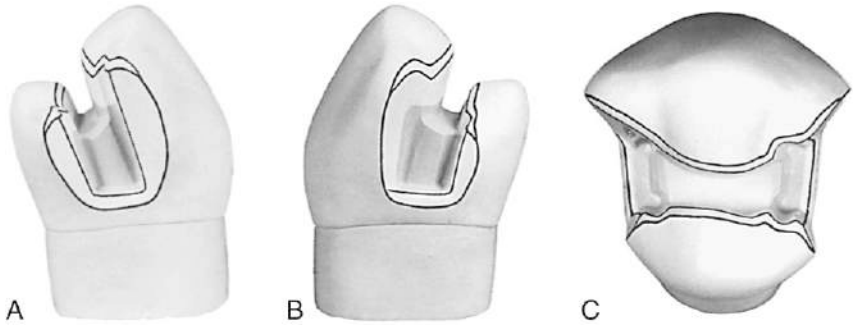


Figure 23.13 Mandibular first premolar prepared for the mesio-occluso-distal inlay. Distal view (A), mesial view (B) and occlusal view (C).

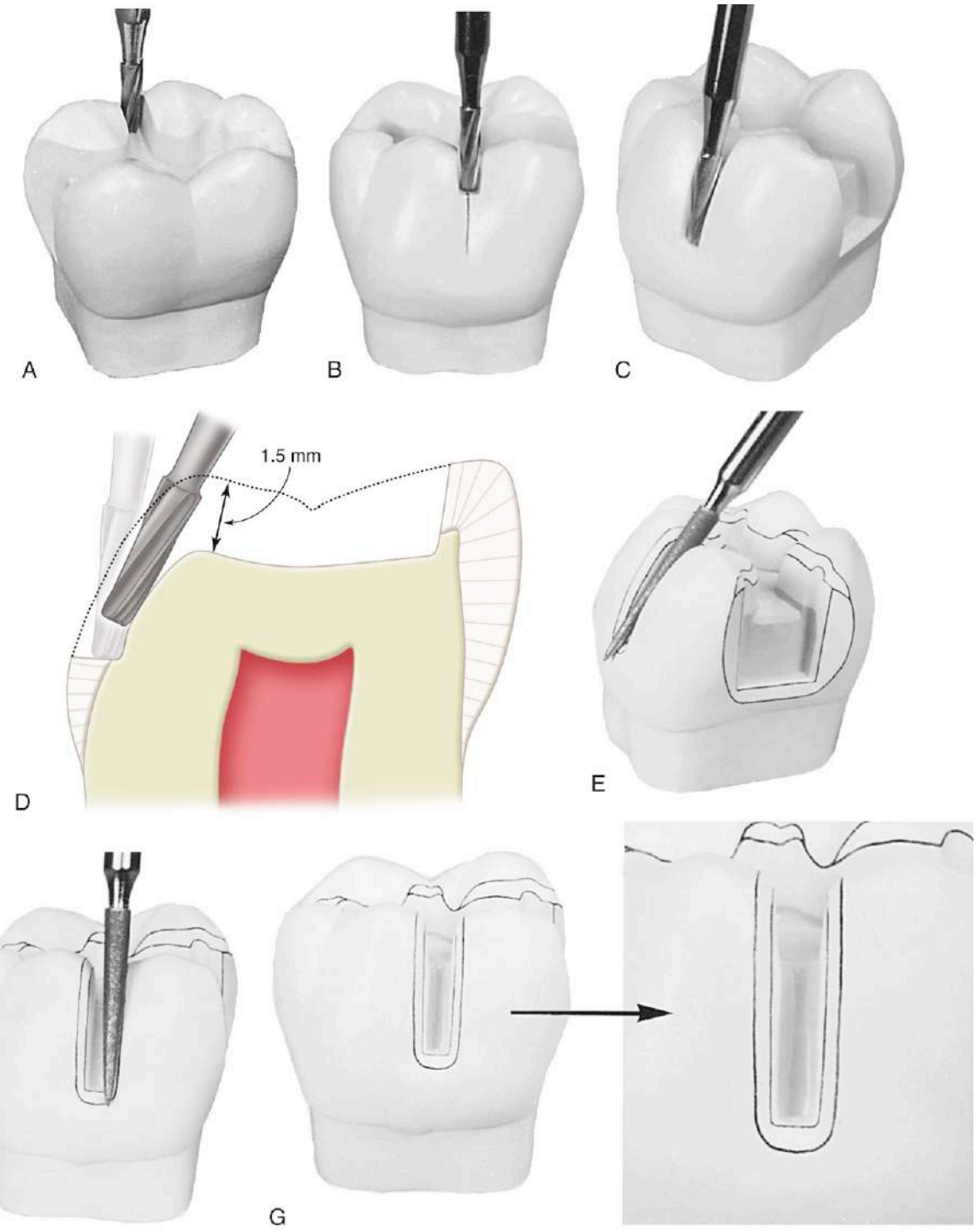


Figure 23.14 (A–C) Extending to include the occlusal fissure that is continuous with the facial fissure on the facial surface. (D) Section of C. (E and F) Bevelling the gingival margin (E) and the mesial and distal margins (F) of fissure extension. (G) Bevelling completed.

bur serves as a depth gauge for the axial wall, which is in dentin. The blade portion of the No. 271 bur is 0.8 mm in diameter at its tip end and 1 mm at the neck; the axial wall depth should approximate 1 mm or slightly more. The bur should be tilted lingually as it is drawn occlusally, to develop the uniform depth of the axial wall (Fig. 23.14D). The same principles apply for the extension of a lingual surface groove.

When a facial or lingual groove is included, it also must be bevelled. With the flame-shaped, fine-grit diamond instrument, the operator bevels the gingival margin (using no more than one-third the depth of the gingival floor) to provide for 30-degree marginal metal (Fig. 23.14D). The operator applies a light bevel on the mesial and distal margins that is continuous with the occlusal and gingival bevels and results in 40-degree metal at these margins (Fig. 23.14F and G). The bevel width around the extended groove is approximately 0.5 mm.

Class II Preparation for Abutment Teeth and Extension Gingivally to Include Root-surface Lesions

Extending the facial, lingual and gingival margins may be indicated on the proximal surfaces of abutments for removable partial dentures to increase the surface area for the development of guiding planes. In addition, the occlusal outline form must be wide enough faciolingually to accommodate any contemplated rest preparation without involving the margins of the restoration. These extensions may be accomplished by simply increasing the width of the bevels.

The following modified preparation is recommended when further gingival extension is indicated to include a root lesion on the proximal surface. The gingival extension should be accomplished primarily by lengthening the gingival bevel, especially when preparing a tooth that has a longer clinical crown than normal as a result of gingival recession. It is necessary to extend (gingivally) the gingival floor only slightly, and although the axial wall consequently must be moved pulpally, this should be minimal. If additional extension of the gingival floor is necessary, it should not be as wide pulpally as when the floor level is at a normal position (Fig. 23.15A). These considerations are necessary because of the draft requirement and because the tooth is smaller apically. Extending the preparation gingivally without these modifications would result in a dangerous encroachment of the axial wall on the pulp (Fig. 23.15B).

Cusp-capping Partial Onlay

Definition. The term *partial onlay* is used when a cast metal restoration covers and restores *at least one but not all* of the cusp tips of a posterior tooth.

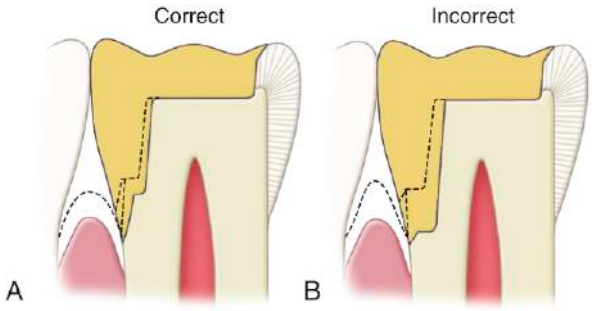


Figure 23.15 Modifications of the preparation when extending to include the proximal root-surface lesions after moderate gingival recession. (A) Correct. (B) Incorrect. Note the decreased dentinal protection of the pulp compared with the management depicted in *A*.

Concept

I. When the occlusal outline is extended up the cusp slopes by *more than half the distance* from any primary occlusal groove (central, facial or lingual) to the cusp tip, capping of the cusp should be *considered*.
II. If the preparation outline is extended *two-thirds the distance* from any primary occlusal groove or more, capping the cusp is usually necessary so as to:
   i. Protect the weak, underlying cuspal structure from fracture caused by masticatory forces.
   ii. Remove the occlusal margin from a region subjected to heavy stress and wear (Fig. 23.16A–C).

The various steps of preparing a partial onlay are illustrated in Fig. 23.16A–L.

Preparation for Full Cast Metal Onlays

Definition. The full *onlay* by definition caps all of the cusps of a posterior tooth and can be designed to help strengthen a tooth that has been weakened by caries or previous restorative experiences.

Concept

- The cast metal onlay restoration spans the gap between the inlay, which is primarily an intra-coronal restoration, and the full crown, which is a totally extra-coronal restoration.
- It can be designed to distribute occlusal loads over the tooth in a manner that greatly decreases the chance of future fracture.[4,5]
- It is more conservative of the tooth structure than the full crown preparation, and its supragingival margins are less irritating to the gingiva.

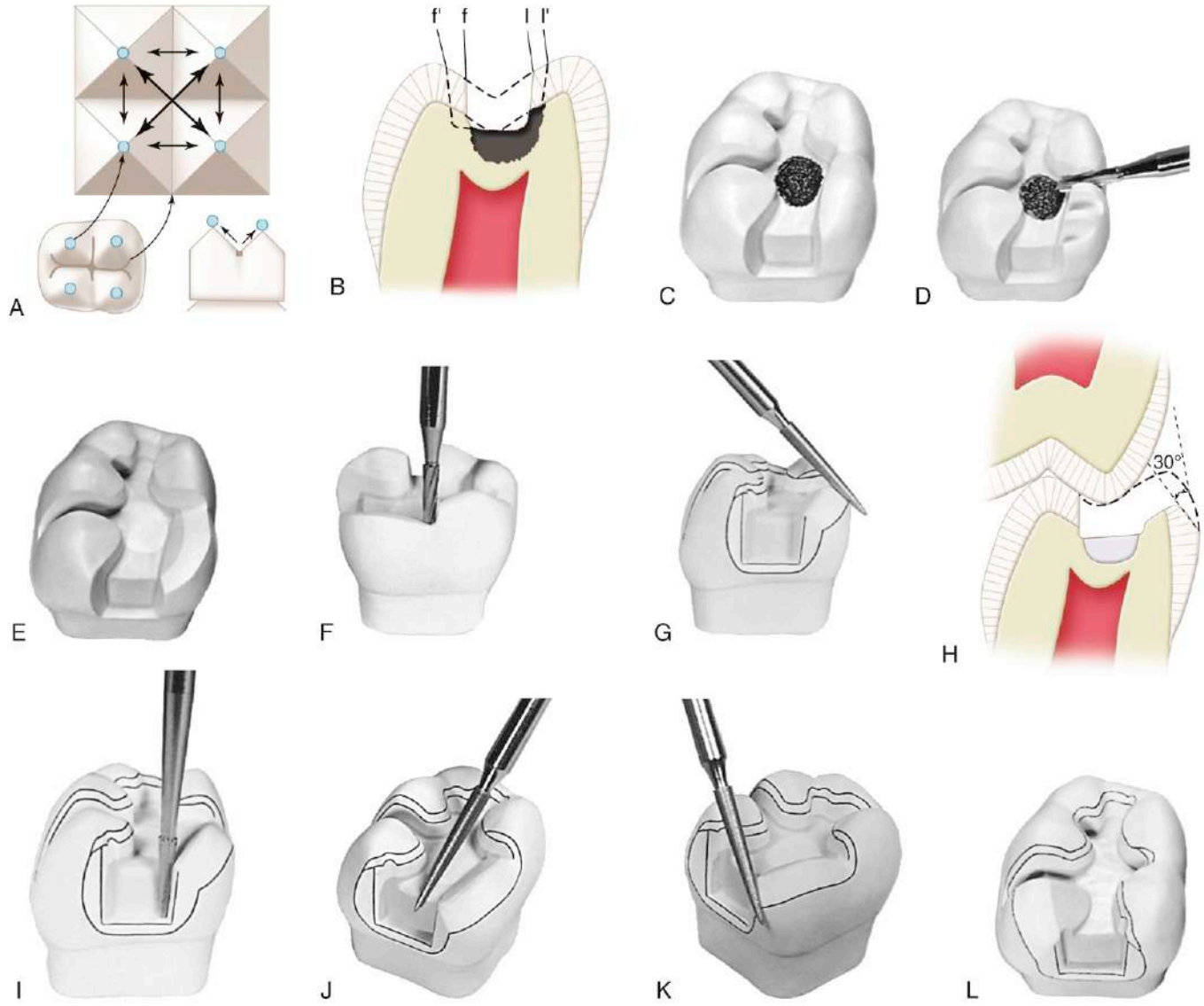


Figure 23.16 (A) When the extension of the occlusal margin is one-half the distance from any point on the primary grooves (*cross*) towards the cusp tip (*dot*), capping of the cusp should be considered; when this distance is two-thirds or more, capping of the cusp is usually indicated. (B) *l* is midway between the central groove and the lingual cusp tip; *f* is midway between the central groove and the facial cusp tip. When enamel at *l* and *f* is undermined by caries, the respective walls must be extended to the dotted lines *l* and *f* to uncover caries. Cusps should be reduced for capping. (C) Extension to uncover caries indicates that the mesiolingual cusp should be reduced for capping. (D) Depth cuts. (E) Reduced mesiolingual cusp. Caries has been removed, and the cement base has been placed. (F) Applying the bur vertically helps establish the vertical wall that barely includes the lingual groove. (G) Counterbevelling reduced cusp. (H) Section of the counterbevel. (I) Improving the retention form by cutting the proximal retention grooves. (J and K) The preparation is complete except for the rounding of the axiopulpal line angle (*J*) and the rounding of the junction of the counterbevel and the secondary flare (*K*). Facial surface groove extension improves the retention and resistance forms. (L) Preparation when reducing one of two facial cusps on the mandibular molar.

The following section would give a stepwise illustration of the preparation of a MOD cast metal onlay in a mandibular first molar. The clinical steps of a cast metal onlay are summarized in Table 23.2.

Step 1: Preparation of Pulpal Floor

- The *No. 271 carbide bur* is oriented continually to a single draw path, usually the long axis of the tooth crown, so that the completed preparation has no undercuts.
- A *2-mm* deep pulpal floor is prepared along the central groove (Fig. 23.17A).
- The occlusal preparation is extended facially and lingually just beyond the caries to sound tooth structure (Fig. 23.17B). The groove should not be extended farther than *two-thirds the distance* from the central groove to the cusp tips because the need for cusp reduction is verified at this point.

Table 23.2 Clinical steps in the preparation of a class II MOD metal onlay

Step 1: Preparation of pulpal floor
Step 2: Preparation of depth gauge grooves
Step 3: Cusp reduction
Step 4: Occlusal step
Step 5: Reverse bevel or counterbevel
Step 6: Proximal box
Step 7: Removal of infected carious dentin and defective restorative materials and pulp protection
Step 8: Preparation of bevels and flares
Step 9: Verification of occlusal clearance

- The gingival-to-occlusal divergence of the preparation walls may range from *2 to 5 degrees* from the line of draw, depending on their heights.
- If the vertical walls are unusually short, a minimum of *2 degrees* occlusal divergence is desirable for retentive purposes.
- As the gingivo-occlusal height of the vertical walls increases, the occlusal divergence should increase, allowing *5 degrees* in the preparation of the greatest gingivo-occlusal length.

Step 2: Preparation of Depth Gauge Grooves

- Before reducing the cusp, the operator prepares depth gauge grooves (depth cuts) with the side of the *No. 271 carbide bur.*
- Uniform *1.5-mm deep* depth cuts are prepared on the remaining occlusal surface (Fig. 23.17C and D).
- Depth cuts usually are placed on the:
  - Crest of the triangular ridges.
  - Facial and lingual groove regions.
- If a cusp is in infraocclusion of the desired occlusal plane before reduction, the amount of cusp reduction is *less* and needs only that which provides the required clearance with the desired occlusal plane.

Step 3: Cusp Reduction

- The reduction should provide for a *uniform 1.5 mm of metal thickness* over the reduced cusp.
- On maxillary premolars and first molars, the reduction should be *minimal* (i.e. *0.75–1 mm*) on the facial cusp ridge to decrease the display of metal (Fig. 23.17A and C).
- The operator should not attempt to reduce the mesial and distal marginal ridges completely at this time to avoid hitting an adjacent tooth. They are reduced in a later step when the proximal boxes are prepared.

Step 4: Occlusal Step

- After cusp reduction, a *0.5-mm deep occlusal step* should be present in the central groove region between the reduced cuspal inclines and the pulpal floor.
- Maintaining the pulpal depth (0.5 mm) of the step, it is extended facially and lingually just beyond any carious areas, to sound tooth structure.
- Next, the operator extends the step mesially and distally far enough to expose the proximal DEJ (Fig. 23.17F).
- The facial and lingual walls of the occlusal step should go around the cusps in graceful curves, and the isthmus should be only as wide as necessary to be in sound tooth structure or sound base or restorative material.
- As the occlusal step approaches the mesial and distal surfaces, it should widen faciolingually in anticipation of the proximal box extensions (Fig. 23.17F). This *0.5 mm occlusal step* contributes to the retention of the restoration and provides the wax pattern and cast metal onlay with additional bulk for rigidity.[6]

Step 5: Reverse Bevel or Counterbevel

Definition. A *reverse bevel* or *counterbevel* is a bevel of generous width, which is prepared on the facial (lingual) margin of a reduced cusp with a flame-shaped, fine-grit diamond instrument.

- The width of the bevel should extend *beyond* any occlusal contact with opposing teeth, either in maximum intercuspation or during mandibular movements.
- It should be at an angle that results in *30 degree marginal metal*.
- The best way to judge this is to always maintain a 30 degree angle between the side of the instrument and the external enamel surface beyond the counterbevel (Fig. 23.18B and C).
- The counterbevel usually should be wide enough so that the cavosurface margin is beyond (gingival to) any contact with the opposing dentition.

Step 6: Proximal Box

Continuing with the No. 271 carbide bur held parallel to the long axis of the tooth crown, the proximal boxes are prepared as described in the inlay section. Fig. 23.17G illustrates the preparation after the proximal boxes are prepared.

Step 7: Removal of Infected Carious Dentin and Defective Restorative Materials and Pulp Protection

If the occlusal step and the proximal boxes have been extended properly, any caries or previous restorative materials remaining on the pulpal and axial walls should

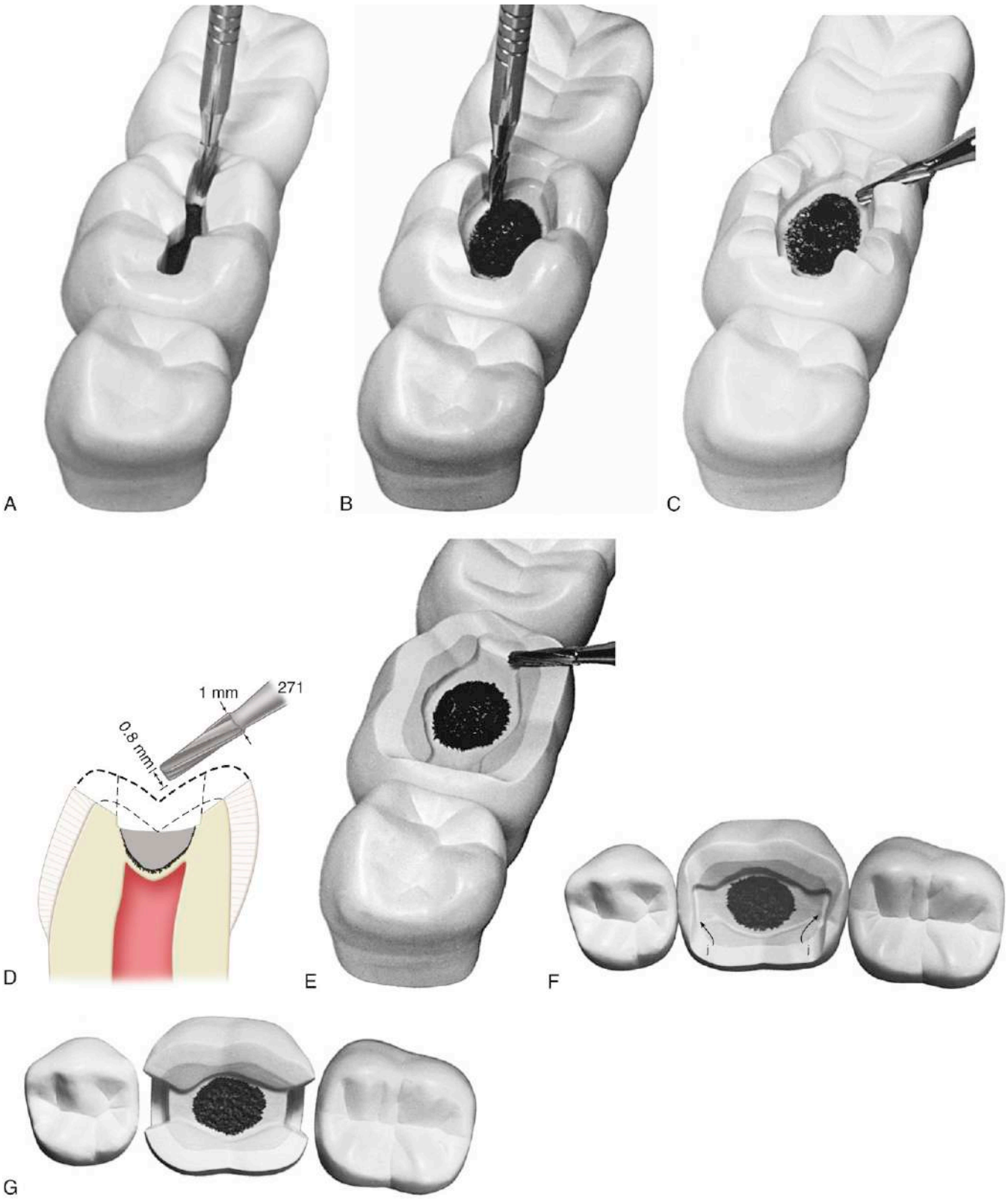


Figure 23.17 (A) Cutting a 2-mm deep central groove. (B) Extending the central groove cut facially and lingually to verify any need for cusp capping. (C) Depth cuts. (D) Section of *C*. (E) Completion of cusp reduction. Small portions of the mesial and distal marginal ridges are left unreduced to avoid scarring the adjacent teeth. (F) The occlusal step is extended facially and lingually past any carious areas and is extended to expose the proximal dentinoenamel junction (DEJ) (*j*) in anticipation of proximal boxing. (G) Preparation with proximal boxes prepared. Note the clearances with the adjacent teeth.

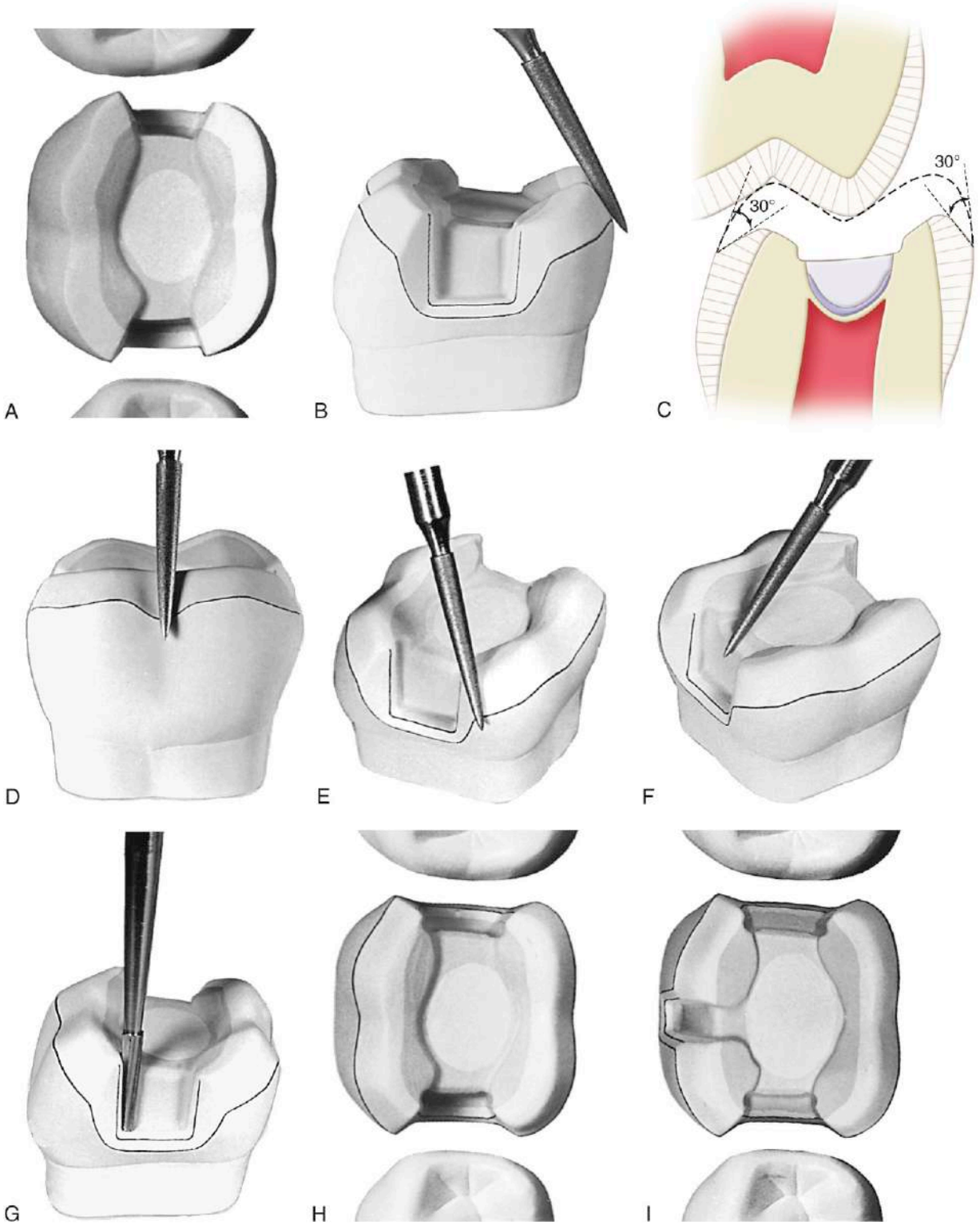


Figure 23.18 (A) Caries has been removed, and the cement base has been inserted. (B) Counterbevelling facial and lingual margins of reduced cusps. (C) Section of *B*. (D) The fissure that extends slightly gingival to the normal position of the counterbevel may be included by slightly deepening the counterbevel in the fissured area. (E) The junctions between the counterbevels and the secondary flares are slightly rounded. (F) The axiopulpal line angle is lightly bevelled. (G) Improving the retention form by cutting proximal grooves. (H) Completed mesio-occluso-distal onlay preparation. (I) Completed mesio-occluso-distofacial onlay preparation showing the extension to include the facial surface groove or fissure.

be visible. They should be removed as described in the inlay section.

Step 8: Preparation of Bevels and Flares

- After the cement base (when indicated) is completed (Fig. 23.18A), the slender, flame-shaped, fine-grit diamond instrument is used to apply the *gingival bevels*, and to create *secondary flares* on the facial and lingual walls of the proximal boxes.
- After bevelling and flaring, any sharp junctions between the counterbevels and the secondary flares are rounded slightly (Fig. 23.18E).
- The fine-grit diamond instrument is also used to bevel the *axiopulpal line angles* lightly (Fig. 23.18F). Such a bevel produces a stronger wax pattern at this critical region by increasing its thickness.
- The desirable metal angle at the margins of onlays is 40 degrees except at the gingivally directed margins, where the metal angle should be 30 degrees.

Step 9: Verification of Occlusal Clearance

- A *wax interocclusal record* is helpful when checking the occlusal clearances, especially in areas that are difficult to visualize, for example, in the central groove and lingual cusp regions.
- To make a wax *bite,* the dentist lightly presses a portion of softened, low-fusing inlay wax over the prepared tooth; the dentist immediately requests the patient to close into the soft wax and slide the teeth in all directions.
- The dentist cools the wax and carefully removes it, holds it up to a light and notes the degree of light transmitted through it. With experience, this is a good indicator of the thickness of the wax.

Enhancement of Resistance and Retention Forms

When the tooth crown is short (which is often the case when all cusps are reduced), the operator must strive to maximize the resistance and retention form in the preparation. The features recommended for enhancing the retention and resistance form are summarized in Table 23.3. These include:

I. **Minimal amount of taper** (2 degrees per wall) on the vertical walls of the preparation
II. Preparation of facial (or lingual) **surface groove extensions**
III. **Proximal retention grooves** (see Fig. 23.16I)
   - When deemed necessary, shallow (*0.3-mm deep*) retention grooves may be cut in the facioaxial and the linguoaxial line angles with the *No. 169L carbide bur* (Fig. 23.18G).
   - These grooves are especially important for retention when the prepared tooth is *short*, which is often the case after reducing all the cusps.
   - The long axis of the No. 169L bur must be held parallel to the line of draw, and the tip of the bur must be positioned in the gingival box internal point angles.
IV. **Skirt Preparation**
   - Definition
     Skirts are thin extensions of the facial or lingual proximal margins of the cast metal onlay that extend from the primary flare to a termination just past the transitional line angle of the tooth.
   - Concept
     – A skirt extension is a conservative method of improving the retention and resistance forms of the preparation.
     – The addition of properly prepared skirts virtually eliminates the chance of postrestorative fracture of the tooth because the skirting onlay is primarily an extra-coronal restoration that encompasses and braces the tooth against forces that might otherwise split the tooth.
     – It is relatively atraumatic to the tooth because the skirt extensions are prepared entirely in enamel.
   - Technique
     The preparation of a skirt is done entirely with the *slender, flame-shaped, fine-grit diamond instrument*.
     Step 1: Maintaining the long axis of the instrument parallel to the line of draw, the operator should create a *definite vertical margin*, just beyond the line angle of the tooth, providing at the same time a 140 degree cavosurface enamel angle (40 degree metal angle) (Fig. 23.19D–F).
     Step 2: Extending into the gingival third of the anatomic crown is usually necessary for an effective resistance form. In most instances, the *gingival margin of the skirt extension is occlusal to the position of the gingival bevel of the proximal box.*
     Step 3: The operator should translate the instrument from the entrance cut towards the proximal box to blend the skirt into the primary flare and the proximal gingival margin (Fig. 23.19G).
     Step 4: Holding the diamond instrument at the same angle that was used for preparing the counterbevel,

Table 23.3 Additional features for enhancing resistance and retention forms in cast restorations

I. Minimal amount of taper (2 degrees per wall) of the vertical walls of the preparation
II. Preparation of facial (or lingual) surface groove extensions
III. Proximal retention grooves
IV. Skirt preparation
V. Collar preparation
VI. Slot preparation

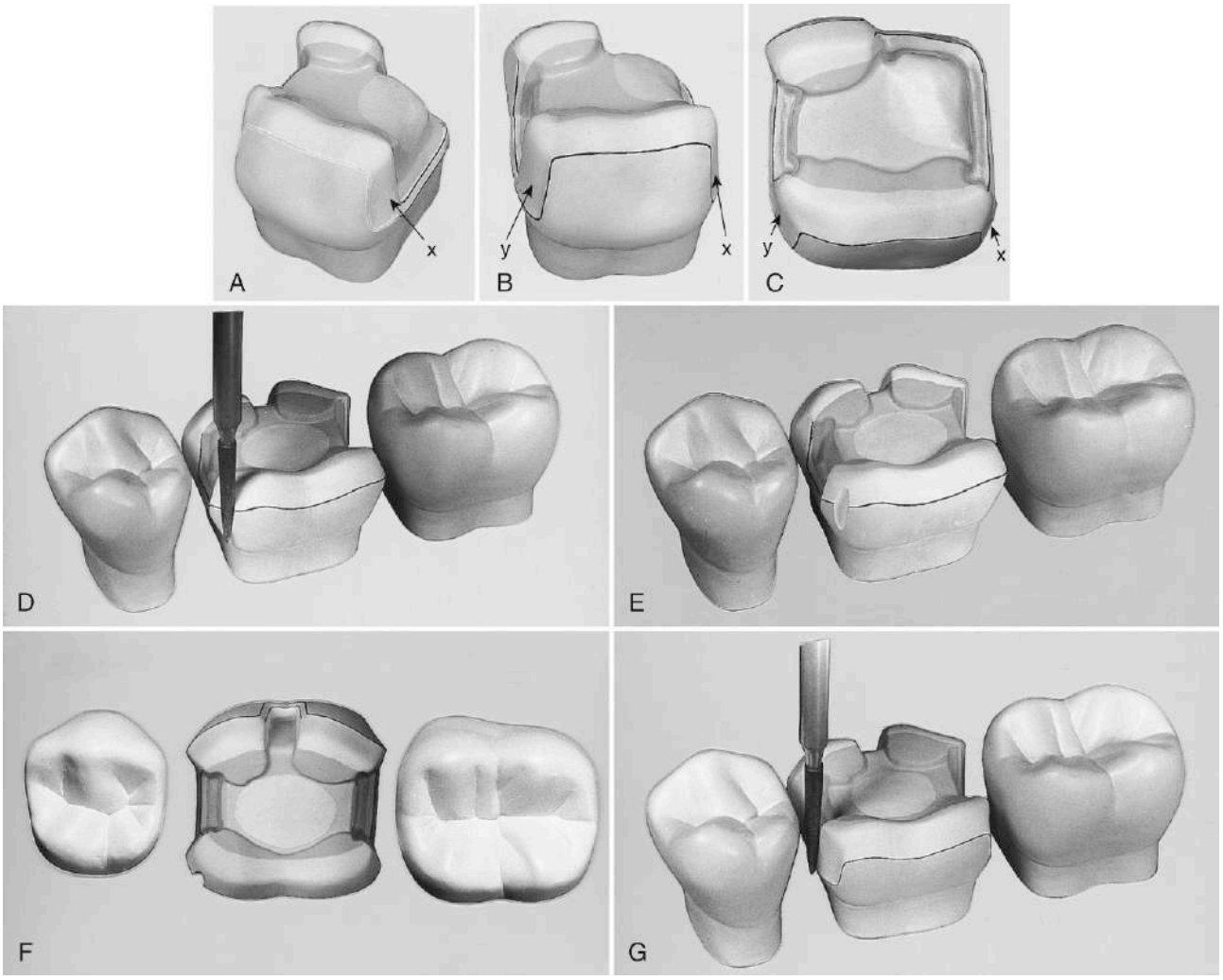


Figure 23.19 (A) When the lingual wall of the proximal box is inadequate or missing, the retention form can be improved by facial skirt extension (*x*). (B) Facio-occlusal view of *A*. Maximal resistance form is developed by skirting the distofacial (*y*) and mesiofacial (*x*) transitional line angles. (C) Occlusal view of *B*. (D–F) The initial cut for the skirt is placed just past the transitional line angle of the tooth. (G) Blending the skirt into the primary flare.

the operator should rounden the junction between the skirt and the counterbevel.

- Clinical Considerations
  - When the proximal portion of a class II preparation for an onlay is being prepared and the lingual wall is partially or totally missing, the retention form normally provided by this wall can be developed with a *skirt extension of the facial margin* (Fig. 23.19A–C).
  - Similarly, if the facial wall is not retentive, a skirt extension of the lingual margin supplies the desired retention form.
  - When the lingual and facial walls of a proximal box are both inadequate, skirt extensions on the respective lingual and facial margins can satisfy the retention and resistance form requirements.
  - The addition of skirt extensions is also recommended when the proximal surface contour and contact are to be *extended more* than the normal dimension to develop a proximal contact.
  - A disadvantage of skirting is that it increases the display of metal on the facial and lingual surfaces of the tooth. For this reason, skirts are not placed on the mesiofacial margin of maxillary premolars and first molars.

v. Collar Preparation

To increase the retention and resistance forms when preparing a weakened tooth for a MOD onlay to cap all cusps, a facial or lingual 'collar' or both may be provided (Fig. 23.20).

- Technique

Step 1: The operator uses a *No. 271 carbide bur* at high speed parallel to the line of draw to prepare a *0.8-mm deep shoulder* (equivalent to the diameter of the tip end of the bur) around the lingual (or facial) surface to provide for a collar about *2–3 mm* high occlusogingivally (Fig. 23.20A and B).

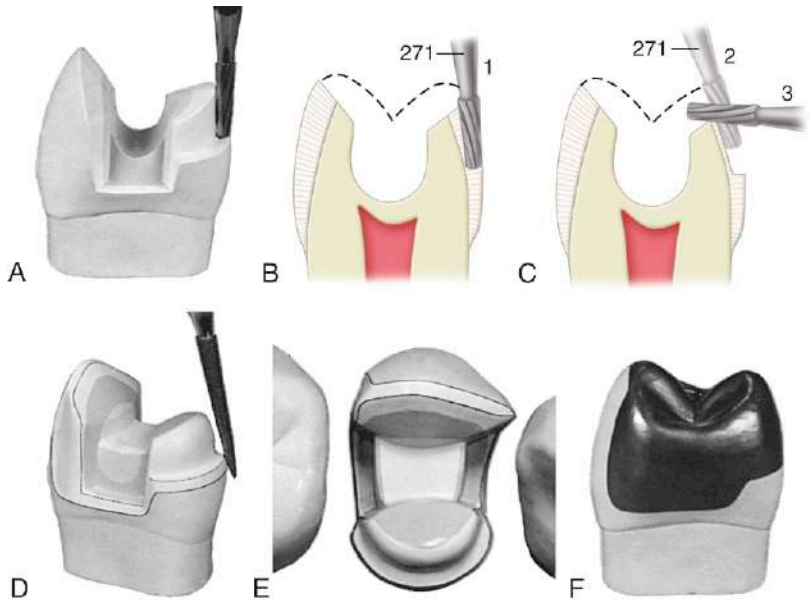


Figure 23.20 (A) First position of the bur in preparing for the lingual collar on a weakened maxillary premolar. (B and C) Section drawings of the first position of the bur (*B*) and the second and third positions (*C*). (D) Bevelling the lingual margin. Note the distofacial skirt extension. (E) Completed preparation. (F) Completed onlay.

Step 2: To provide for a uniform thickness of metal, the occlusal 1 mm of this reduction should be prepared to follow the original contour of the tooth (Fig. 23.20C).

Step 3: Any undesirable sharp line angle formed by the union of the prepared lingual and occlusal surfaces should be rounded. This aspect of the preparation is completed by lightly bevelling the gingival margin of the shoulder with the *flame-shaped, fine-grit diamond instrument* to achieve a 30 degree metal angle at the margin (Fig. 23.20D).

- The lingual 'skirt' extension, the lingual 'collar' preparation or the lingual surface groove extension on a maxillary molar protects the facial cusps from fracture.
- The facial skirt extension, the facial collar preparation or the facial surface groove extension on a mandibular molar protects the lingual cusp from fracture.

VI. Slot Preparation

Occasionally, the use of a slot in dentin is helpful in creating the necessary retention form (Fig. 23.21A–C). A slot is preferred over preparing a box because:

- A slot conserves more of the tooth structure.
- The linear extent of marginal outline is lesser for a slot.

▪ Technique

- To form this slot, the dentist uses a *No. 169L carbide bur* with its long axis parallel to the line of draw (this must be reasonably close to a line parallel with the long axis of the tooth) (Fig. 23.21A).
- The slot is cut in dentin so that it would pass midway between the pulp and the DEJ if it were to be extended gingivally (Fig. 23.21C).
- The slot should have the following approximate dimensions:
  - width (diameter) of the bur mesiodistally (0.5 mm),
  - length of 2 mm faciolingually and
  - depth of 2 mm gingival of the normally positioned pulpal wall.

▪ Advantages

The position and direction of the slot helps in:

- preventing the exposure of the pulp,
- minimizing the removal of the dentin supporting the enamel and
- preventing the perforation of the distal surface of the tooth at the gingival termination of the slot.

Restorative Techniques for Cast Metal Restoration

The restorative technique for a cast metal restoration can be divided into the following stages as shown in Table 23.4.

Interocclusal Record

The maximum inter-cuspation interocclusal record can be made from one of several commercially available bite registration pastes. The most commonly used bite registration pastes are composed of *heavily filled PVS impression materials*. Several materials are available in cartridge systems that automatically mix the base and accelerator pastes together

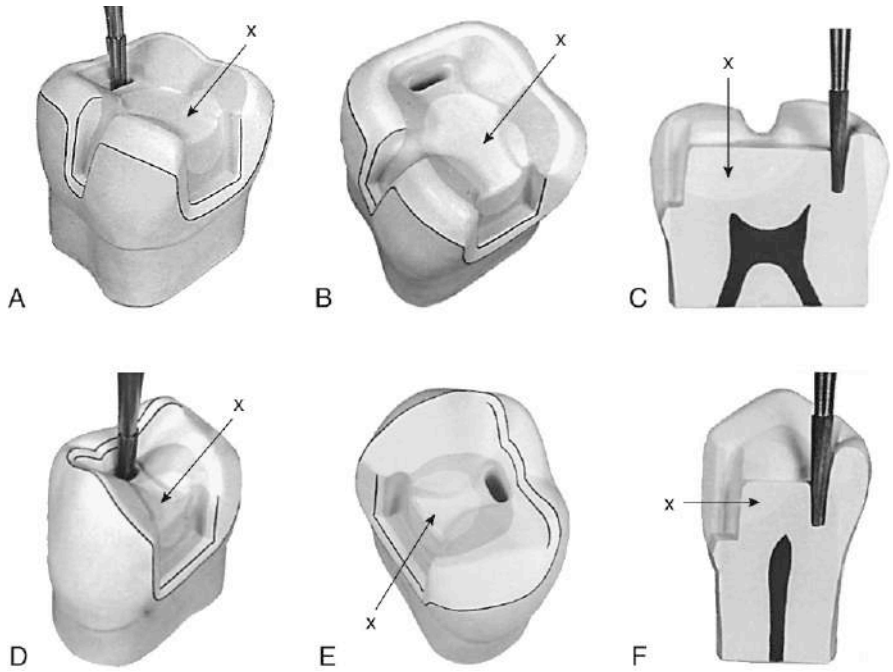


Figure 23.21 (A and B) Cutting a distal slot for the retention for the mesio-occlusal onlay to treat the terminal molar having a large cement base (*x*) resulting from extensive occlusal and mesial caries. (C) Section of *A*. (D and E) Preparing a mesial slot for the retention for the disto-occlusal onlay to treat the maxillary first premolar that has a large cement base (*x*). (F) Section of *D*.

as they are expressed through a special disposable mixing tip (Fig. 23.22A).

Technique

- The mixed impression material is dispensed directly onto the prepared teeth and their opponents, and then the patient closes the mouth completely (Fig. 23.22B and C).
- The dentist observes teeth not covered by the bite registration paste to verify that teeth are in maximum inter-cuspation.

Table 23.4 Stages of laboratory fabrication of a cast metal restoration

I. Interocclusal record
II. Custom temporary restoration
III. Final impression
  i. Tissue retraction
  ii. Polyvinyl siloxane (PVS) impression
    – Tray selection
    – Impression technique
IV. Working cast and dies
V. Wax pattern
VI. Spruing, investing casting, finishing, seating, adjusting and polishing the casting
VII. Trying in the casting
  i. Preparing the mouth
  ii. Seating the casting and adjusting proximal contacts
  iii. Occluding the casting
  iv. Improving marginal adaptation
VIII. Cementation

- When the material has set, the dentist removes the interocclusal record and inspects it for completeness. When held up to a light, areas where the adjacent unprepared teeth have penetrated through the material should be seen (Fig. 23.22D).

Custom Temporary Restoration

Between the time the tooth is prepared and the cast metal restoration is delivered, it is important that the patient be comfortable and the tooth be protected and stabilized with an adequate temporary restoration. The custom temporary restoration should satisfy the following requirements:

I. It should be *nonirritating* and protect the prepared tooth from injury.
II. It should *protect* and maintain the health of the periodontium.
III. It should maintain the *position* of the prepared, adjacent and opposing teeth.
IV. It should provide for *esthetic*, *phonetic* and *masticatory* function, as indicated.
V. It should have adequate strength and retention to withstand the forces to which it will be subjected.

Temporaries can be fabricated by two methods:

I. **Direct technique**: Intra-orally directly on the prepared teeth.
II. **Indirect technique**: Extra-orally outside of the mouth using a postoperative cast of the prepared teeth. The indirect technique is not as popular as the direct technique because of the increased number of steps and complexity in the former.

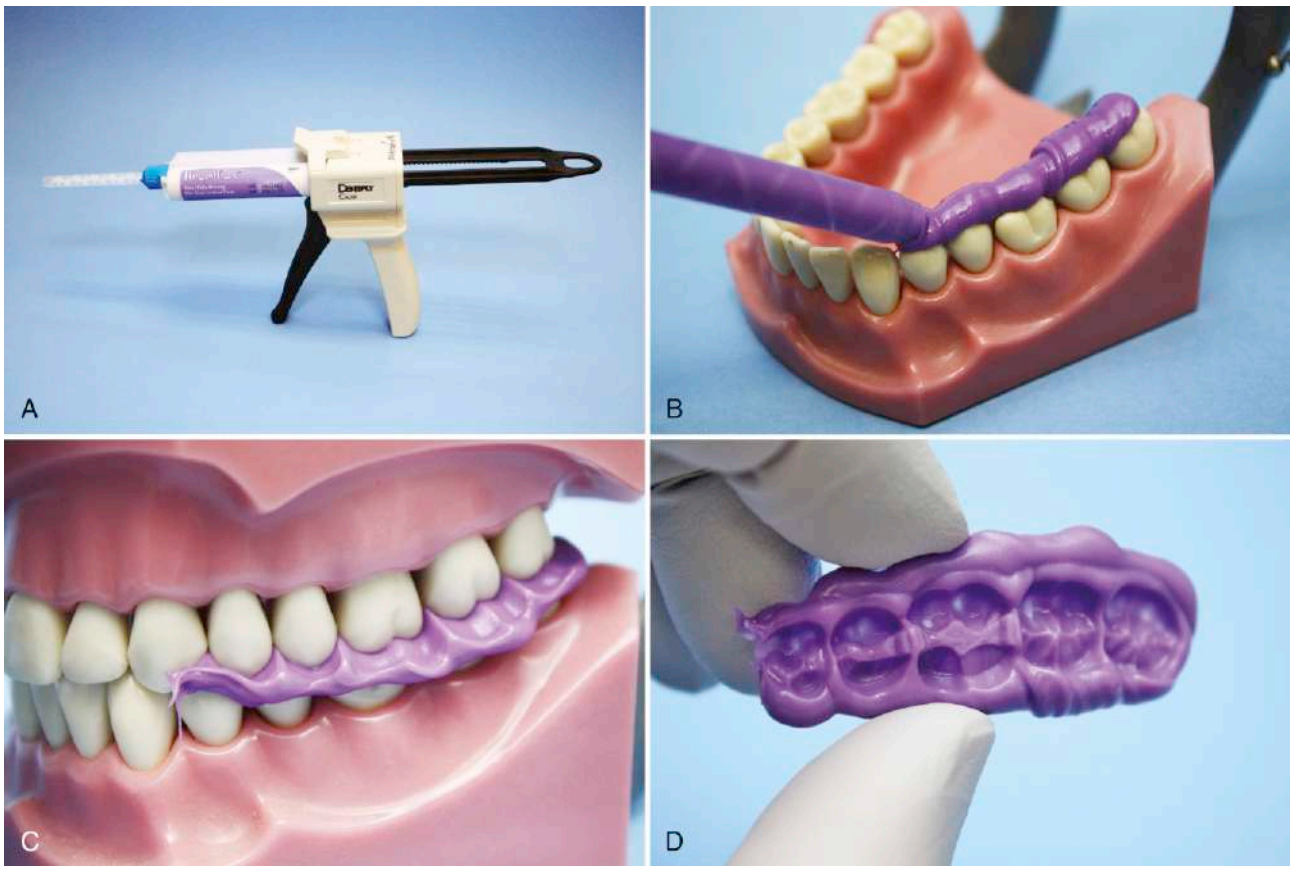


Figure 23.22 Maximum inter-cuspation interocclusal record made with polyvinyl siloxane bite registration paste. (A) One of many commercially available bite registration materials used in this technique. (B) Using a cartridge dispenser and a disposable automixing tip, the base and accelerator pastes are automatically mixed and applied to the prepared teeth, their neighbours and the opposing teeth. (C) Have the patient close into maximum inter-cuspation position. Be sure that the adjacent unprepared teeth are touching in their normal relationships. (D) Remove the maximum inter-cuspation interocclusal record carefully after it has set, and inspect it for completeness. Areas where the adjacent, unprepared teeth have penetrated through paste should be seen.

Technique for Direct Temporary Restoration

The direct temporary technique involves forming the temporary restoration directly on the prepared tooth (Fig. 23.23).

Advantages

I. The direct technique involves fewer steps and materials because no postoperative impression and gypsum cast are required.
II. It is much faster than the indirect technique.

Disadvantages

I. There is a chance of locking hardened temporary materials into small undercuts on the prepared tooth and the adjacent teeth.
II. The marginal fit may be slightly inferior to the indirect technique.
III. It is more difficult to contour the temporary restoration without the guidelines offered by the postoperative cast.[7]

Technique

Step 1: Forming the temporary restoration directly on the prepared tooth requires the *preoperative impression*.
Step 2: Trial-fitting seats the preoperative impression onto teeth to verify that it seats completely.
Step 3: The temporary material is mixed, following the manufacturer's instructions. Temporary materials that use automixing tips are especially convenient (Fig. 23.23C). The dentist places the material into the preoperative impression in the area of the prepared tooth, taking care not to entrap any air (Fig. 23.23D).
Step 4: The impression is placed on teeth, and the dentist ensures that it seats completely (Fig. 23.23E). Most temporary systems recommend monitoring the setting by rolling some excess material into a small ball and holding it between two fingers.
Step 5: When the temporary material has set to a firm stage, the impression is removed. The formed temporary restoration should remain on the prepared tooth.
Step 6: The operator tests the temporary restoration by pressing on the occlusal surface slightly, and when the

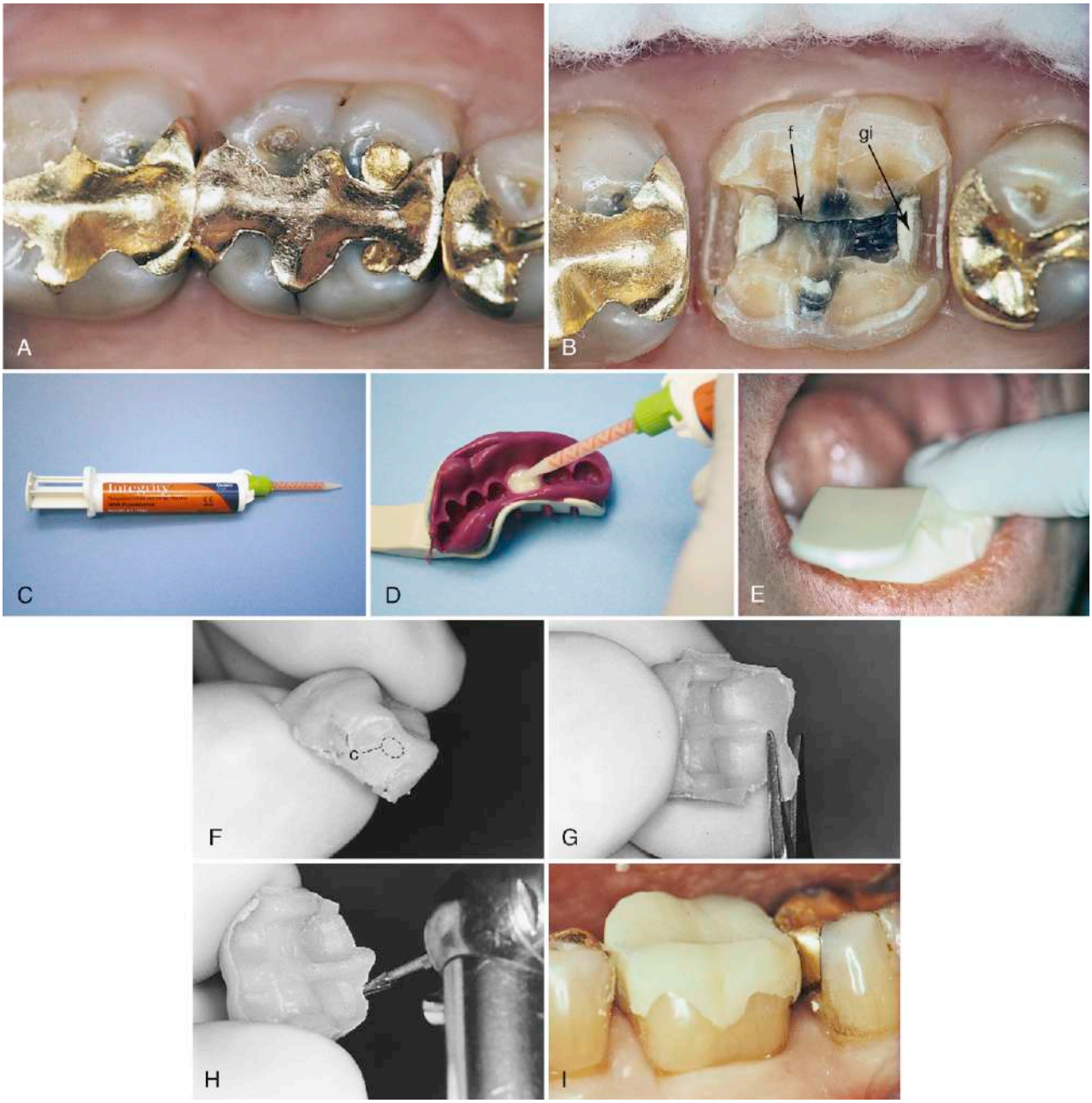


Figure 23.23 Forming the direct temporary restoration with preoperative impression. Mesio-occluso-disto-facial onlay preparation for the mandibular first molar is used for illustration. (A) Preoperatively, this patient had symptoms indicating an incomplete fracture of a vital tooth. (B) After preparation for an onlay, an incomplete fracture (*f*) of dentin is seen extending mesiodistally along the pulpal floor. To maximize the retention and resistance forms, all the cusps are reduced for capping, a facial surface groove extension is prepared and all four transitional line angles have skirt extensions. Glass ionomer cement bases are inserted into excavations on axial walls (*gi*). Cement bases should have a light coat of water-soluble lubricant to prevent adhesion. (C) Automixing the temporary resin material. (D) The mixed temporary material is poured into the preoperative impression of the prepared tooth. (E) The preoperative impression is seated with the temporary material onto the prepared tooth. (F) The formed temporary restoration is removed from the preparation (note the contact area *c*, which must not be removed during trimming). (G) Thin excess can be removed by using scissors. (H) The internal surface of the temporary restoration has record of the cavosurface margin that is used as guide for final trimming. (I) After the final impression is made, the temporary restoration is cemented with temporary cement. The temporary material over the skirt extensions is left slightly overcontoured for additional strength.

material is sufficiently strong, the operator removes it from the tooth (Fig. 23.23F).

Step 7: Excess material is trimmed away (Fig. 23.23G). The cavosurface margins of the preparation can be seen inside the temporary restoration and are used as a guide for trimming the critical external areas near the margins (Fig. 23.23H).

Step 8: Trial-fit the temporary restoration on the patient's teeth. It should fit well, make desirable contact with the adjacent teeth and meet occlusal requirements with minimal adjustments.

Step 9: If occlusal adjustments are indicated, the high points are marked with articulating paper and reduced with an appropriate rotary instrument.

Step 10: The temporary restoration is removed from the mouth and set aside for cementation with temporary cement after the final impression has been made (Fig. 23.23I).

Final Impression

The impression material used for the final impression of a cast metal restoration should have the following qualities:

I. It must become *elastic* after placement in the mouth because it must be withdrawn from undercut regions that usually exist on the prepared and adjacent teeth.
II. It must have adequate *strength* to resist breaking or tearing on removal from the mouth.
III. It must have adequate *dimensional accuracy, stability* and *reproduction of detail* so that it is an exact negative imprint of the prepared and adjacent unprepared teeth.
IV. It must have handling and *setting characteristics* that meet clinical requirements.
V. It must be free of toxic or irritating components.
VI. It must be possible to disinfect it without distorting it.

The most common impression material used for the indirect casting technique is *PVS*. The technique for the use of this material is discussed in detail in the following sections.

The final impression would incorporate the following two stages:

I. Tissue retraction
II. PVS Impression
   a. Tray Selection
   b. Impression Technique

Tissue retraction

Final impression materials make accurate impressions only of tooth surfaces that are visible, clean and dry. When margins are subgingival, it is necessary to use *retraction cords* to displace the free gingiva temporarily away from the tooth and to control the flow of any gingival haemorrhage and sulcular fluids. The objective of gingival retraction is to:

I. *Widen the gingival sulcus* to provide access for the impression material to reach the subgingival margins in adequate bulk to resist tearing during impression withdrawal.
II. *Control of haemorrhage and moisture* by the use of retraction cord impregnated with appropriate *styptics* (e.g. aluminum chloride), *vasoconstrictors* (e.g. epinephrine) or both.

The use of vasoconstrictors in retraction cord is contraindicated in some patients, especially those who have cardiac arrhythmias, severe cardiovascular disease, uncontrolled hyperthyroidism or diabetes and patients taking drugs such as b-blockers, monoamine oxidase inhibitors or tricyclic antidepressants.[8]

Technique

Step 1: The dentist selects and cuts a retraction cord of suitable diameter that is slightly longer than the length of the gingival margin.

Step 2: Using the edge of a paddle-tipped instrument or the side of an explorer, one end of the cord is gently placed into the sulcus, about 2 mm facial to the point where the facial margin passes under the free gingiva.

Step 3: Then, the cord is inserted progressively into the remainder of the sulcus, with the end of the cord left exposed, to be grasped with tweezers later in the technique.

- Occasionally, when the gingival margin is deep, it is helpful to insert a second cord of the same or larger diameter over the first (Fig. 23.24A and B).
- When the free gingiva is thin and the sulcus is narrow (e.g. facial surface of the maxillary or mandibular canine), a cord of very small diameter must be selected to prevent undue trauma to the tissue.

Polyvinyl Siloxane Impression

The PVS impression is discussed in detail here because it is widely used, and the technique for its use can be readily applied to most other impression materials. PVS impression materials have the following advantages over other impression materials used for final impressions:

I. They have excellent reproduction of detail and dimensional stability over time.
II. They are *user-friendly* because they are easy to mix and have no unpleasant odour or taste.
III. PVS impressions can withstand *disinfection routines* without significant distortion.
IV. These impression materials come in the form of two pastes (*base and catalyst*) that are mixed in disposable, automix, cartridge-dispensing systems. These automix systems provide excellent mixing of the accelerator and base pastes (Fig. 23.25).

Tray selection and preparation

The impression tray must be sufficiently rigid to avoid deformation during the impression technique. Two types of trays are suitable:

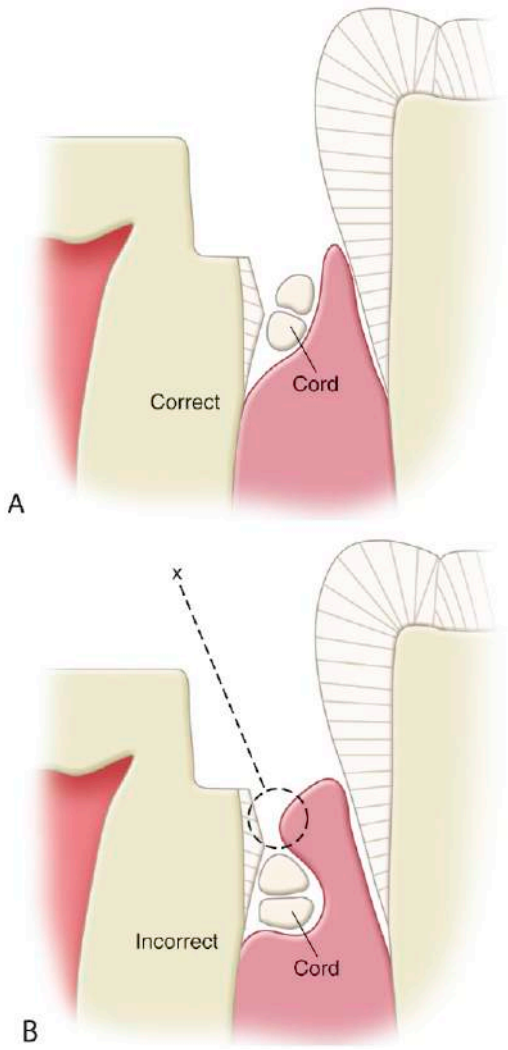


Figure 23.24 (A) Correct application of the retraction cord. (B) Incorrect application of the retraction cord causing the impression material to tear at *x*.



Figure 23.25 Light-bodied (low viscosity), medium-bodied and heavy-bodied (high viscosity) polyvinyl siloxane impression material; dispenser; automixing tip and putty (very high viscosity). (Courtesy: Kerr Corp., Orange, CA).

I. Commercial stock
II. Custom made
- Use of stock plastic trays is convenient and saves time.
- A thickness of impression material greater than 3 mm increases shrinkage and the chance of voids; a thickness less than 2 mm may lead to distortion or tear of the impression material.
- Adequate bonding of impression material to the tray is accomplished with the application of a special adhesive to the tray.

Impression technique

Most dental manufacturers offer their PVS impression materials in automix dispensing systems. The automixing systems have many advantages:

I. Speed
II. Consistent and complete mixing of accelerator and base pastes
III. Incorporation of very few air voids during mixing and delivery to teeth

The technique demonstrated requires *two viscosities* of impression material:

I. A *light-bodied material* to inject around the preparation.
II. A *heavy-bodied material* to fill the tray.

The dispensers are loaded with cartridges that contain the accelerator and base pastes (Fig. 23.26A and B). A disposable automixing tip fits onto the end of each cartridge (Fig. 23.26C). The light-bodied mixing tip has an accessory curved tip that is small enough to gain access to the smallest, most remote areas of the preparation (Fig. 23.26E).

Technique

Step 1: The first dispenser is used to mix and fill the impression tray with the heavy-bodied impression material (Fig. 23.26D). The dispensing tip should be kept embedded in the impression material as it is expressed into the tray so that the chance of trapping air is decreased.
Step 2: Teeth should be examined to ensure that the field is still clean and dry. Any visible moisture on teeth is removed with compressed air. The second dispenser is then used to mix and inject the light-bodied impression material on the prepared teeth (Fig. 23.26E).
Step 3: The retraction cord is gently removed with operative pliers.
Step 4: The opened gingival sulci and preparations are deliberately and progressively (moving from distal to mesial) filled over and beyond the margins with material from the syringe. Light-bodied material also is injected on the occlusal surfaces of the unprepared adjacent teeth to eliminate the trapping of air on the occlusal grooves.
Step 5: After filling and covering teeth with material from the syringe, the cotton rolls are immediately removed and the loaded tray seated over the region till the complete setting of the material.

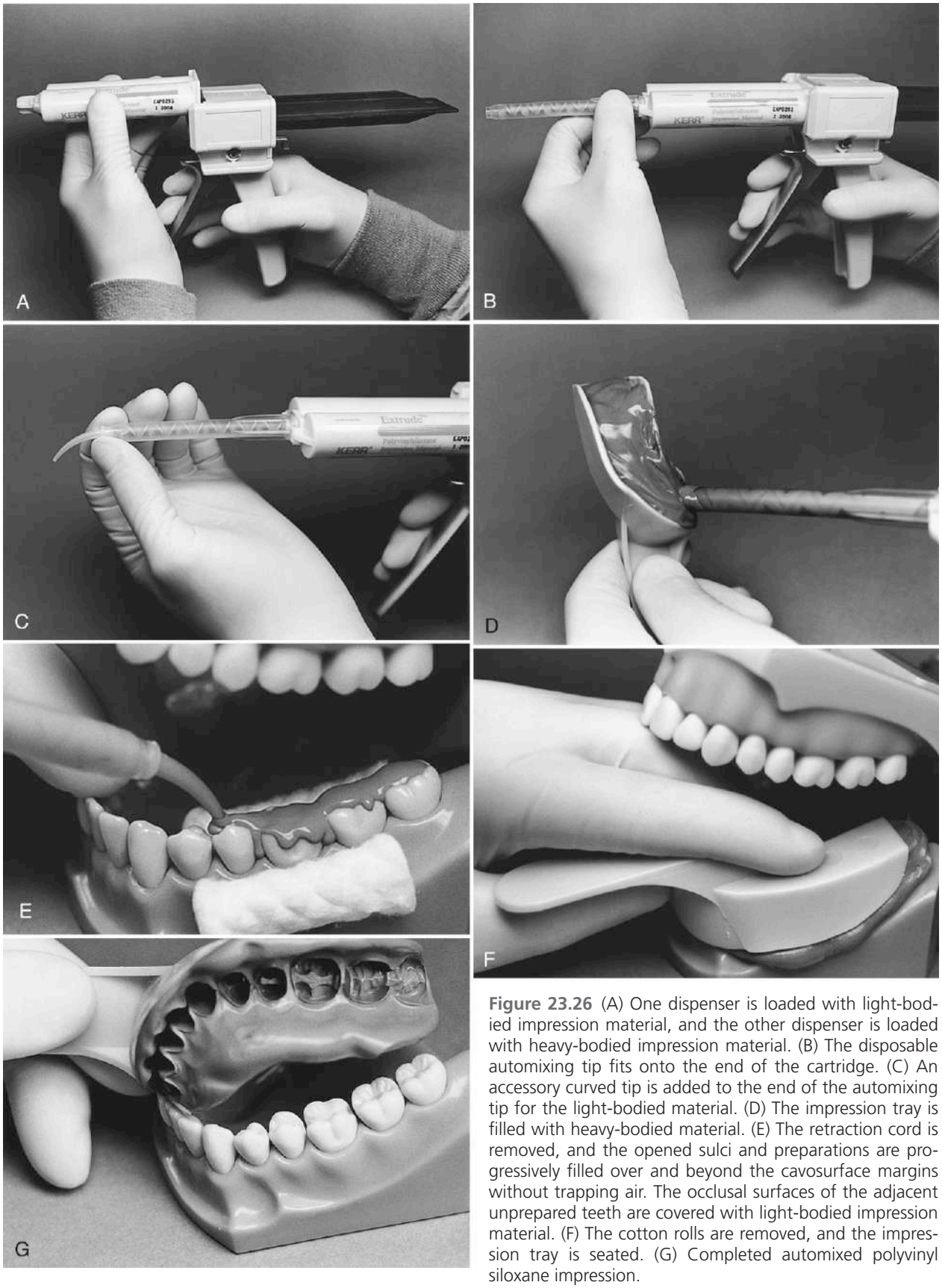


Figure 23.26 (A) One dispenser is loaded with light-bodied impression material, and the other dispenser is loaded with heavy-bodied impression material. (B) The disposable automixing tip fits onto the end of the cartridge. (C) An accessory curved tip is added to the end of the automixing tip for the light-bodied material. (D) The impression tray is filled with heavy-bodied material. (E) The retraction cord is removed, and the opened sulci and preparations are progressively filled over and beyond the cavosurface margins without trapping air. The occlusal surfaces of the adjacent unprepared teeth are covered with light-bodied impression material. (F) The cotton rolls are removed, and the impression tray is seated. (G) Completed automixed polyvinyl siloxane impression.

Step 6: After the PVS impression has properly polymerized, it is removed from the mouth by a quick, firm pull that is directed as much as possible in line with the draw of the preparation. Removal is aided by inserting a fingertip at the junction of the facial border of the impression and the vestibule fornix, disrupting the vacuum that occasionally occurs during withdrawal, especially with full-arch impressions.

Working Casts and Dies

Definitions

I. **Working cast:** The *working cast* is an accurate replica of the prepared and adjacent unprepared teeth that allows the cast metal restoration to be fabricated in the laboratory.
II. **Dies**: Dies are replicas of the prepared teeth and the adjacent unprepared teeth that are individually removable to help in the fabrication procedure of cast metal restorations.

Technique

The most used methods for creating a working cast with removable dies from an elastic impression require *two pours*. The first pour is made to produce the removable dies, and the second pour is made to establish intra-arch relationships. Working casts made in this manner are called *split casts*. Several satisfactory methods are available for making a split cast with removable dies. The **Pindex system** (Coltene/Whaledent Inc, Cuyahoga Falls, OH) is illustrated because it offers many advantages, as follows:

I. The first pour becomes the die segment and can be made quickly and easily.
II. Dowel pins can be positioned precisely, where needed.
III. Dowel pins are automatically positioned parallel, which facilitates die removal.

Fig. 23.27 illustrates the step-wise method for making a split cast with removable dies using the Pindex system.

Wax Patterns

Technique

Step 1

The operator lubricates the die and incrementally adds liquid wax from a No. 7 wax spatula by the 'flow and press' method to form the proximal, facial and lingual surface aspects of the pattern (Fig. 23.28A and B).

Step 2

The ability to form proper proximal contour and contact (Fig. 23.29) is the advantage of an indirect cast restoration over a direct restoration like amalgam. Drawings of two maxillary premolars (Fig. 23.29) are used to illustrate forms of contact and mesiodistal widths of interproximal spaces.

Normal contact

In Fig. 23.29A, the position of the contact is marked with an *x*, the area of near-approach of the two surfaces is indicated with a broken line and the position of the crest of the gingiva is indicated with a continuous line. Fig. 23.29B, is a mesiodistal section through teeth at the point of contact, and Fig. 23.29C, is an occlusal view.

Broad contact

Broad faciolingual contact A broad contact faciolingually is illustrated in Fig. 23.29D–F. In the proximal view (Fig. 23.29D), the position of a normal contact is marked *x*, the contact of this tooth is the outlined oblong area and the area of near approach is the broken line. The crest of the gingiva is less arched, being almost horizontal along the area of near-approach. Viewed from the facial in the mesiodistal section (Fig. 23.29E), the contact appears to be the same as in Fig. 23.29B, but a comparison of the occlusal views (Fig. 23.29F and C) shows the extra breadth of this contact at the expense of the lingual embrasure.
Broad occlusogingival contact A contact that is too broad in the occlusogingival direction, but narrow faciolingually, is illustrated in Fig. 23.29K and L. The principal objections to this form of contact are that stringy foods which are likely to be caught and held; also, if proximal recurrent caries occurs, it is farther to the gingival, requiring a tooth preparation close to the cementoenamel junction (CEJ).

Gingival contact

Fig. 23.29H shows a contact that is too far to the gingival. Its position in comparison with normal is shown by the relation of the circle to the *x* in Fig. 23.29G. The problem with such a contact is in the inclinations of the proximal surfaces from the occlusal marginal ridges to the contact. Stringy food is likely to become packed into this space, and the contact may impinge on the interproximal tissue.

Occlusal contact

Fig. 23.29I and J, for an illustration of a contact too close to the occlusal. This form is frequently observed in restorations (especially amalgams), seldom in virgin teeth. Such a contact allows food to fill the gingival embrasure and invites proximal caries.

Open contact

Total lack of a proximal contact is often referred to as an *open contact* and is to be avoided. Lack of a proximal contact is usually undesirable because it creates the risk of proximal drifting of teeth, shifting occlusion, food impaction and damage to the supporting tissues.

Step 3

Complete the occlusal wax-up with a warmed No. 7 wax spatula (Fig. 23.30A–E). This warming of the spatula

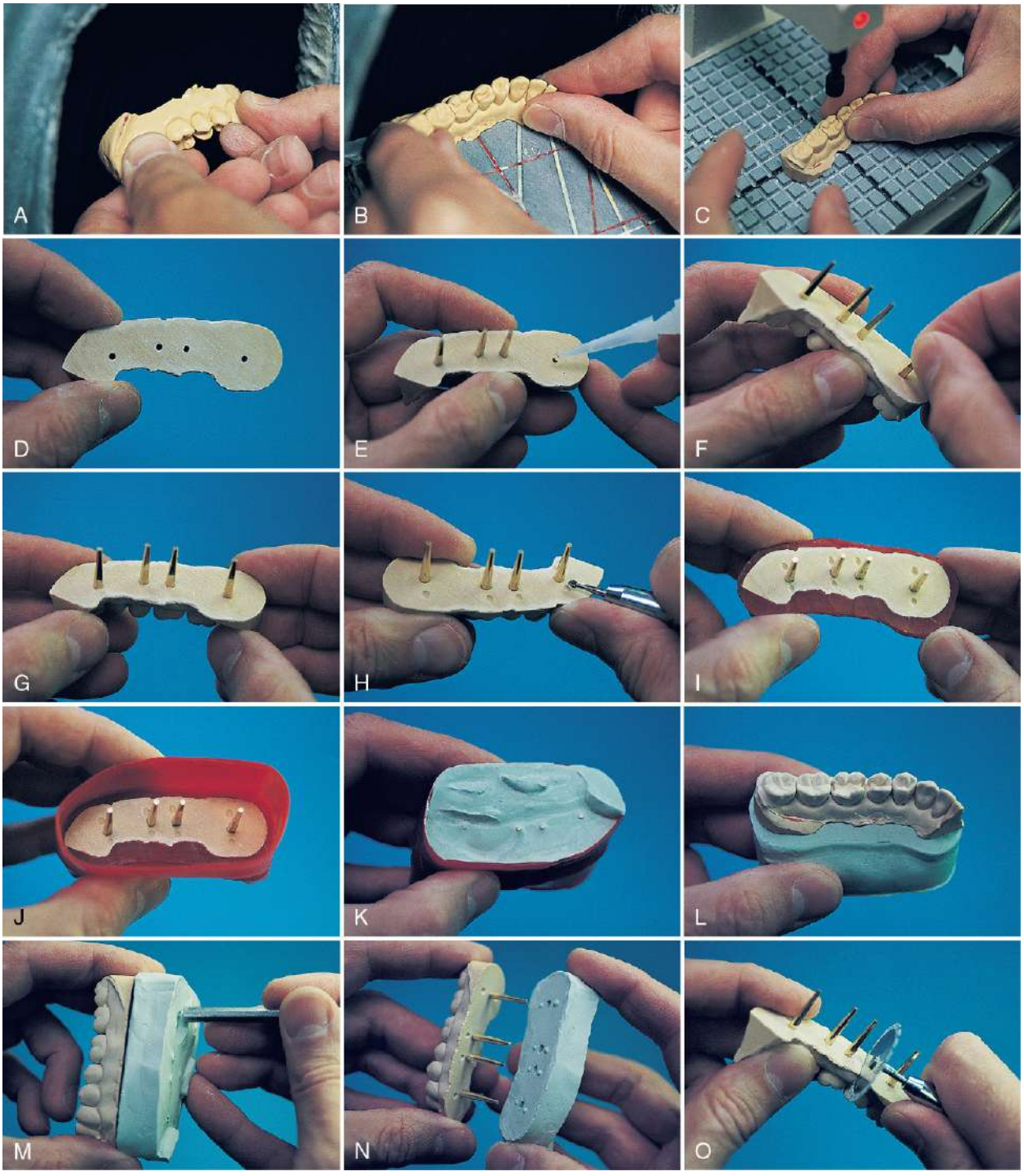


Figure 23.27 (A) The base of the die segment is trimmed flat and approximately parallel to the occlusal surfaces with a model trimmer. Dies should be approximately 15 mm high occlusogingivally. (B) The die segment is trimmed on the facial and lingual surfaces to reduce the need for trimming in later steps. (C) The die segment on the Pindex machine, ready to drill hole for first molar die. A small red dot of light helps position the cast. (D) Holes drilled for removable dies. (E) A drop of cyanoacrylate glue is poured into each hole. The cast must be dry for the glue to adhere. (F) Immediately insert a dowel pin into the hole, being sure it is fully seated. (G) The dowel pins must be parallel to one another and fully seated, and no excess glue must be present. (H) To aid in indexing, small dimples are cut in the base of the dies, using one-third the diameter of a large (No. 6) round bur. Typically, these are positioned facial and lingual to the dowel pin. (I) Rope wax is placed around the cast, flush with the die bases. (J) Boxing wax is placed around the rope wax to create a container for the base pour. A separating agent must be painted on the die bases to prevent adherence to the base pour.(K) Base pour is completed. At least 1 mm of the dowel pin should be left protruding. (L) Cast after removing boxing and rope wax. (M) Tapping on the end of each dowel pin until the die segment moves. (N) Removing the die segment from base. (O) The dies are carefully cut apart by using a saw, bur or thin diamond abrasive disk. Eye protection and dust collection are essential.

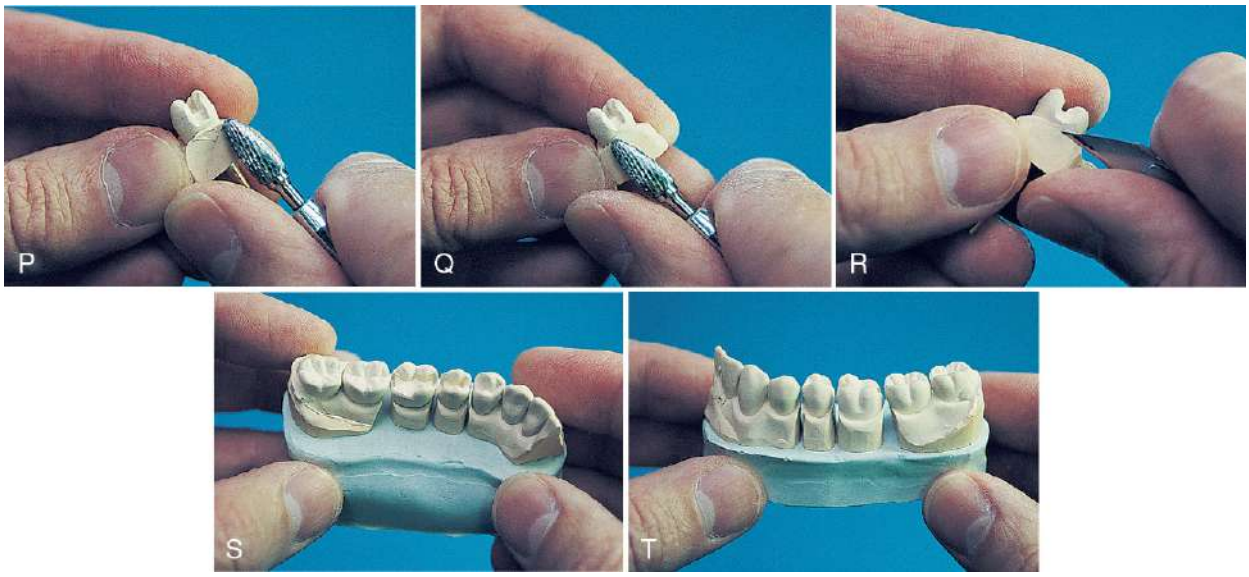


Figure 23.27 (cont.) (P) Excess die stone around the gingival margins usually prevents good access for later steps in fabrication. (Q) Removing the excess die stone with a large crosscut carbide bur in a slow-speed handpiece. Trimming across slightly gingival of the recorded gingival contour of the tooth weakens the excess, causing it to fall away. (R) Final trimming is completed with a sharp scalpel. (S) Cast completed, lingual view. Note how each prepared tooth and the adjacent unprepared teeth are removable now. (T) Cast completed, facial view. Note the full seating of the dies.

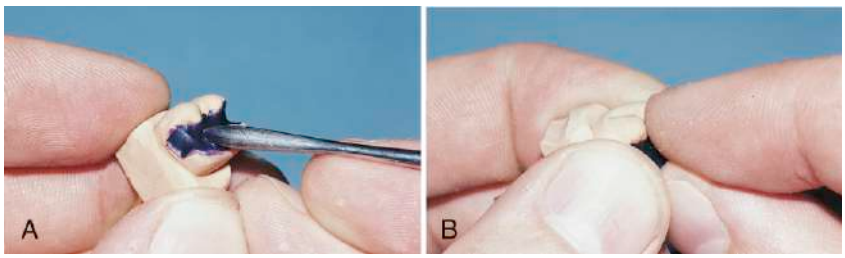


Figure 23.28 To ensure optimal wax adaptation to the preparation walls, a thin layer of wax is first poured (A) and then finger pressure is applied for several seconds while the wax cools (B).

permits carving of the marginal wax with light pressure so that the stone margins are not damaged.

Spruing, Investing, Casting, Finishing, Seating, Adjusting and Polishing the Casting

The reader is referred to textbooks on dental materials for the principles and techniques of spruing, investing, casting, finishing, seating, adjusting and polishing the casting.

Trying-in the Casting

Preparing the Mouth

The temporary restoration is removed, ensuring that all the temporary cement has been dislodged from the preparation walls and cleared away. To improve visualization, the region is isolated with cotton rolls.

Seating the Casting and Adjusting the Proximal Contacts

The operator confirms the fit of the casting on the tooth. A 3 × 3 in. (7.5 × 7.5 cm) gauge sponge should be placed as a *throat screen* to catch the casting if it is accidentally dropped (Fig. 23.31). The dentist tries the casting on the tooth, using light pressure. Adjustment of the proximal contacts is done keeping the following points in mind:

- *Do not force the casting on the tooth.*
- If the casting does not seat completely, the most likely cause is an *overcontoured* proximal surface.
- Using the mouth mirror, where needed, one views into the embrasures from the facial, lingual and occlusal aspects.
- Passing *dental floss* through the contact indicates tightness and position, helping the trained operator identify the degree of excess contact and its location.

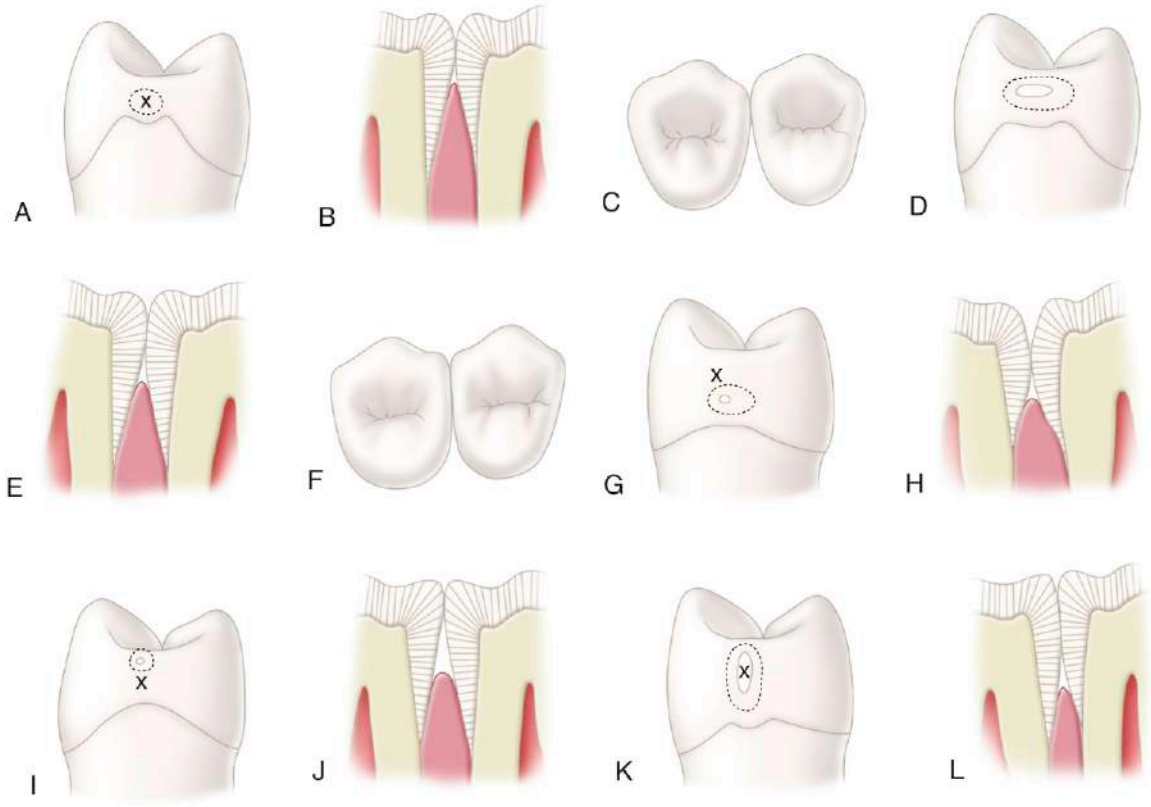


Figure 23.29 (A–C) Correct contact. Note the position and form of the contact and the form of the embrasures around the contact. The mesial and distal pits are below (gingival of) the proximal marginal ridges. (D–F) Contact too broad faciolingually. (G and H) Contact positioned too far gingivally. (I and J) Contact too close to the occlusal surface. (K and L) Contact too broad occlusogingivally.
(Modified from Black GV: *Operative dentistry*, ed 8, Woodstock, IL, 1947, Medico-Dental, vol. 2).

- When adjusting a MOD restoration, *only one* excess contact should be adjusted at a time (the stronger one).
- A *rubber wheel abrasive* is used to adjust the proximal contour and to correct the contact relationship.
- The correct 'tightness' of the contacts is best judged with dental floss.

Occluding the Casting

Inter-cuspal contact

I. The patient is asked to close into *maximum inter-cuspation*, and the dentist inspects the unprepared adjacent teeth to see if any space exists between the opposing wear facets.
II. After drying the teeth of saliva, the dentist inserts a strip of *articulating paper* and requests the patient to close and tap the teeth together (in maximum inter-cuspation) several times.
III. The dentist removes the paper and examines it by holding it up towards the light for evidence of any areas of penetration caused by the restoration. Any holes can be matched with heavy markings on the casting, and *shiny, metal-coloured spots* may be present in the centre of the marks.
IV. Such heavy contacts should be reduced with suitable *abrasive stones*, while carefully observing the fundamental concepts for equilibration of occlusion.

The use of articulating paper and the stone is continued until:

- The heavy markings are no longer produced
- The contacts on the restoration have optimal position and form
- An even distribution of contacts exists on the casting and the adjacent teeth

Lateral working contact

Lateral working (functional) contacts on the casting are marked by:

Step 1: Inserting a strip of articulator paper over the quadrant with the casting.
Step 2: Having the patient close into maximum inter-cuspation.
Step 3: *Sliding* the teeth towards the side of the mouth where the casting is located.

Improving Marginal Adaptation

Step 1: With a *ball* or *beaver-tail burnisher*, the operator improves marginal adaptation by burnishing the marginal

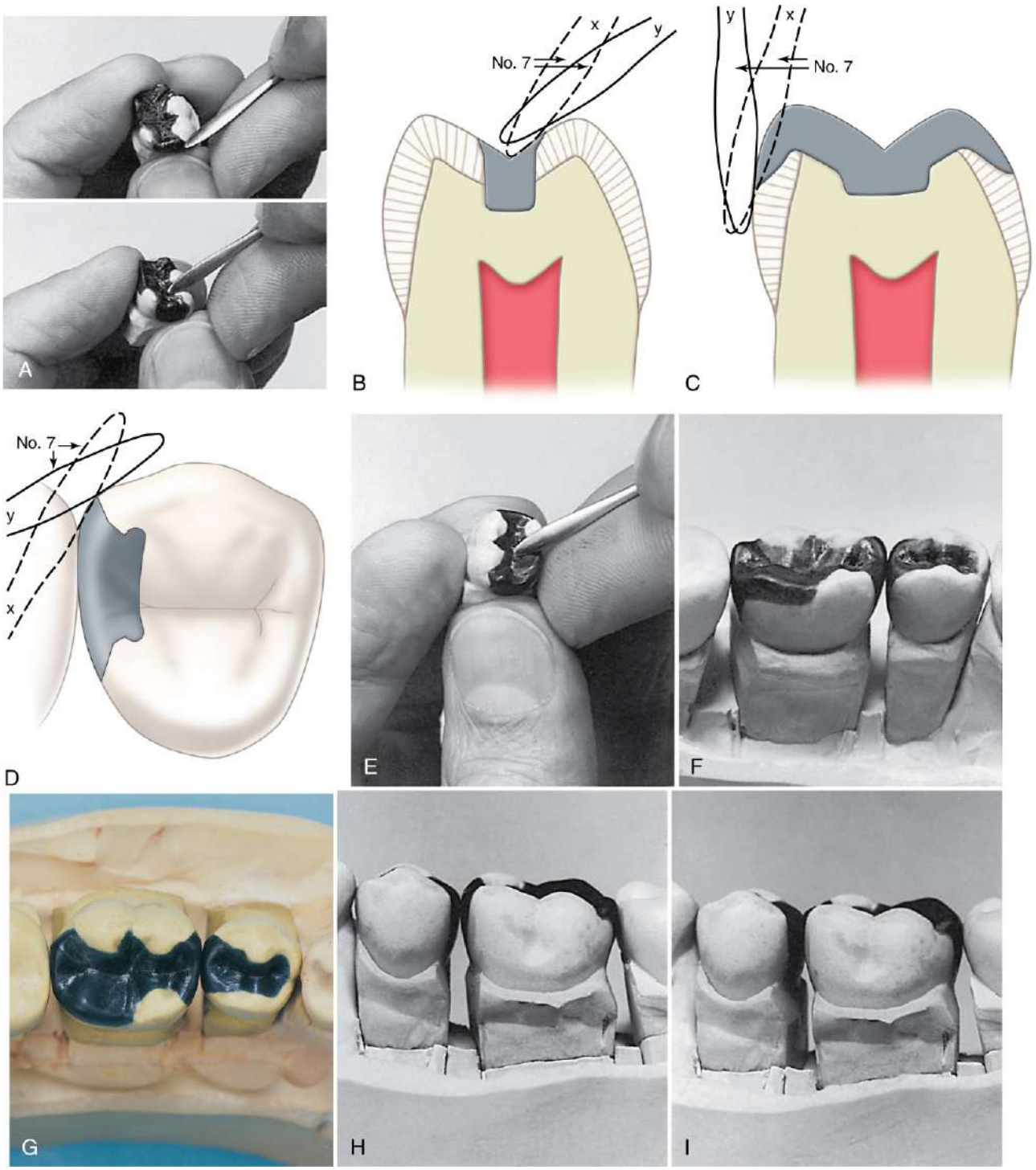


Figure 23.30 (A) Wax is carved to margins with a warm No. 7 spatula. (B–D) Incorrect application of No. 7 spatula to carve the contour of the marginal wax is shown by *x*; the correct manner is labelled *y*. (E) Carving the occlusal groove and pit anatomy. (F) The adjacent marginal ridges should be on the same level as much as possible. (G) Occlusal view of completed patterns. Note the shape of the facial and lingual embrasures and the position of the contact. (H and I) Facial view of the completed patterns. Note the gingival and occlusal embrasures and the position of the contact.

metal with strokes that parallel the margin except for the gingival margin (Fig. 23.32A).

Step 2: If necessary, the marginal adaptation and continuity can be improved further by the application of a *pointed, fine-grit carborundum stone* (Fig. 23.32C).

Step 3: Another instrument that can be used to improve marginal fit in accessible areas (e.g. the occlusal two-thirds of the proximal margins) is a *fine-grit paper disk* (Fig. 23.32D).



Figure 23.31 Place 3 × 3 in. (7.5 × 7.5 cm) gauze throat screen to prevent swallowing or aspiration of casting should it be accidentally mishandled.

- The margins should now be such that the explorer tip can pass across the margins smoothly without jumping or catching.
- The operator should use rubber polishing points of increasing fineness at low speed to smooth and polish the accessible areas of roughness left from adjusting procedures (Fig. 23.32E–G).

Cementation

Step 1: Before cementing the casting, the tooth is isolated from saliva with the aid of cotton rolls (and saliva ejector, if necessary) (Fig. 23.33A).

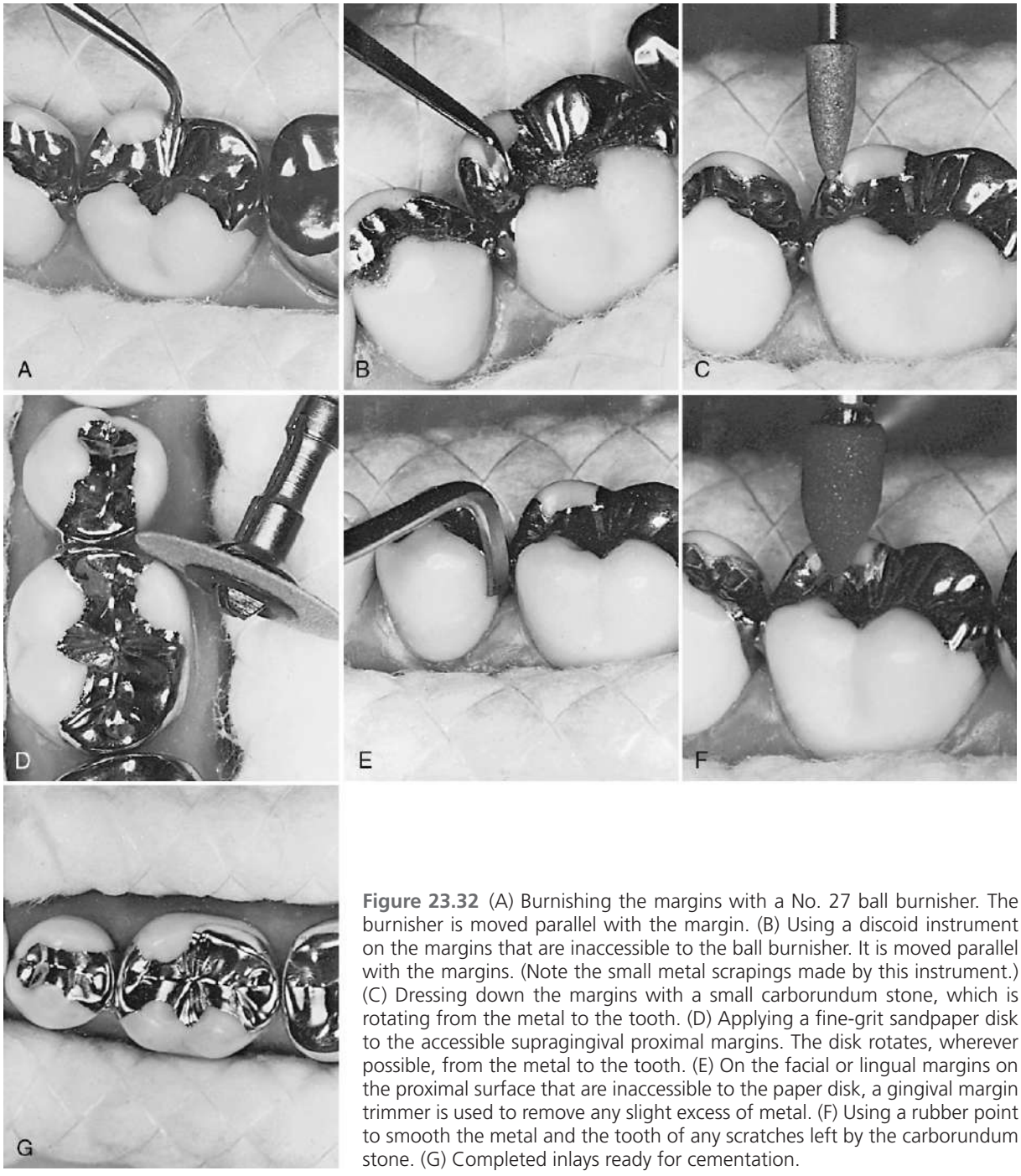


Figure 23.32 (A) Burnishing the margins with a No. 27 ball burnisher. The burnisher is moved parallel with the margin. (B) Using a discoid instrument on the margins that are inaccessible to the ball burnisher. It is moved parallel with the margins. (Note the small metal scrapings made by this instrument.) (C) Dressing down the margins with a small carborundum stone, which is rotating from the metal to the tooth. (D) Applying a fine-grit sandpaper disk to the accessible supragingival proximal margins. The disk rotates, wherever possible, from the metal to the tooth. (E) On the facial or lingual margins on the proximal surface that are inaccessible to the paper disk, a gingival margin trimmer is used to remove any slight excess of metal. (F) Using a rubber point to smooth the metal and the tooth of any scratches left by the carborundum stone. (G) Completed inlays ready for cementation.

Step 2: The cement is mixed according to the manufacturer's instructions and applied generously to the preparation side of the casting (Fig. 23.33B).
Step 3: The dentist starts to place the casting with the fingers or with operative pliers. Next, the dentist places the ball burnisher in the pit areas (first one and then another), exerting firm pressure to seat the casting (Fig. 23.33C).
Step 4: The dentist places a small flexible rubber polishing disk over the casting, removes the saliva ejector and requests the patient to close and exert biting force (Fig. 23.33D and E). The patient also is asked to move

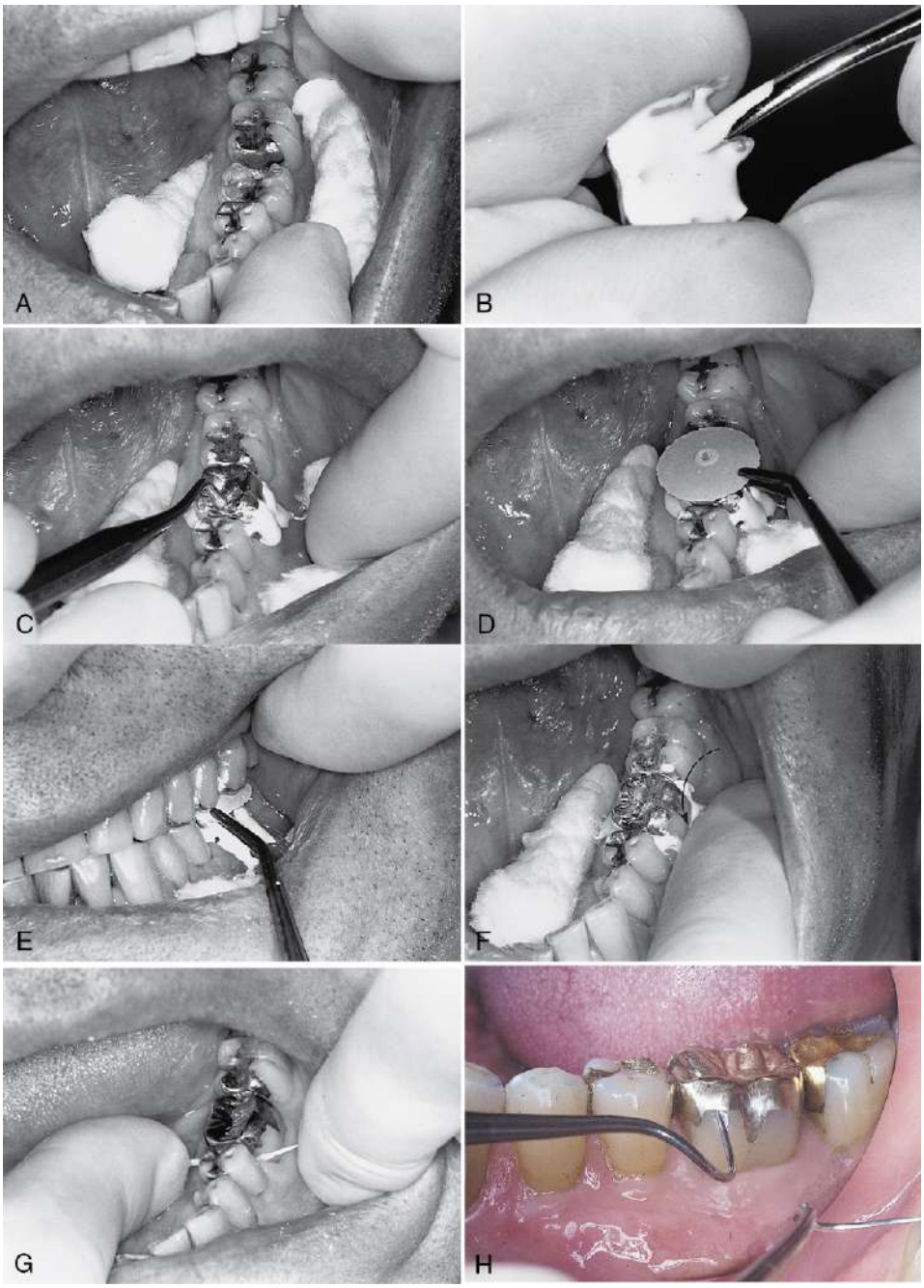


Figure 23.33 Cementing the cast metal onlay on the preparation. (A) Isolating the tooth from saliva with cotton rolls. (B) Applying cement with No. 2 beaver-tail burnisher to preparation side of onlay. (C) Seating the onlay by using a ball burnisher and hand pressure. (D) Placing a rubber polishing disk over the onlay and cementing the cast metal onlay on the preparation. (E) The patient is instructed to apply masticatory pressure while slightly moving the jaw from side to side. (F) When the disk is lifted from the casting, much of the occlusal aspect is free of cement. With a sweeping, rolling motion of the forefinger, any accessible facial surface margin is cleaned of excess cement to permit visual inspection for verification of proper seating of the onlay. Similarly, any accessible lingual margin is cleaned of excess cement. Full seating also should be verified tactilely with the explorer tine. (G) Excess set cement is removed by using the explorer and air–water spray. Dental tape with a small knot is used to dislodge small pieces of interproximal cement. (H) Onlay after cementation.

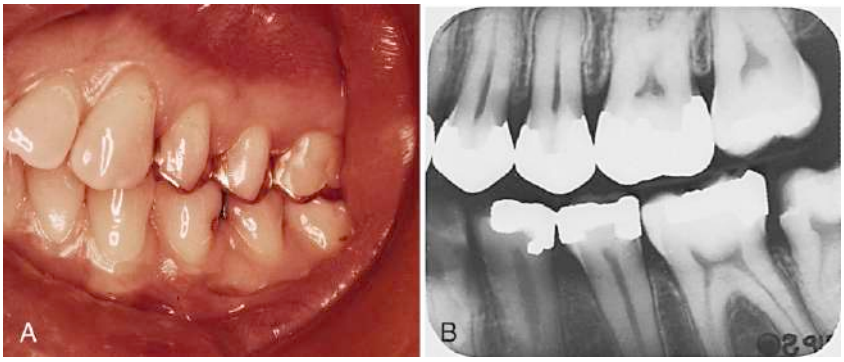


Figure 23.34 (A) Cemented castings. This photo was taken immediately after cementation. (B) Bitewing radiograph of the restored quadrant shown in *A*. Note the fit of inlays at the gingival margins and the contour of the proximal surfaces.

the mandible slightly from side to side, while continuing to exert pressure. A few seconds of this pressure is sufficient.

Step 5: When the disk is removed, much of the occlusal area should be clean of the cement mix and easier to inspect and to verify complete seating of the casting. While the cement is still soft, all accessible margins are burnished. The saliva ejector is replaced in the mouth and the region is kept dry during the setting of the cement.

Step 6: After the cement has hardened, any excess is cleaned off with an explorer and air–water spray. Dental floss should be passed through the contact, carried into the interproximal gingival embrasures and sulci and pulled facially and lingually to help in the removal of cement in this region (Fig. 23.33G). When cementing has been properly accomplished, a cement line should not be visible at the margins (Fig. 23.33H).

A quadrant of inlays after cementation is illustrated in Fig. 23.34.

Summary

Cast metal technique requires multiple patient visits and excellent laboratory support, but the resulting restorations are durable and long lasting. Cast metal onlays, in particular, can be designed to strengthen the restored tooth while conserving more tooth structure than does a full crown. Disadvantages such as high cost and esthetics limit their use, but when indicated, cast metal inlays and onlays provide a restorative option that is less damaging to pulpal and periodontal tissues compared with a full crown.

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